

DIAL: Position-Debiased LLM Judges with Adaptive Human Preference Calibration

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Abstract

Large language models (LLMs) as a judge enable scalable evaluation, but their judgments can be sensitive to response order and, even after removing such position effects, can still diverge systematically from human preferences. We introduce DIAL, a unified framework that combines abundant LLM comparisons with limited human comparisons to separate judge-specific position effects, learn shared structure in position-debiased LLM preferences, and adaptively calibrate that structure toward the human preference target. Theoretically, we study three aspects of DIAL: (i) identification of latent LLM preferences, position effects, and human calibration; (ii) adaptive estimation that balances LLM anchoring against limited human evidence; and (iii) fixed-weight uncertainty quantification for the calibrated human preference. Empirically, we evaluate position debiasing and human alignment separately in controlled simulations and on three human-preference benchmarks, showing that DIAL remains robust to unbalanced response order, achieves strong human-aligned rankings with limited labels, and adapts toward human evidence when LLM information is imperfect. Our real-data study collects over 410K judgments from 21 LLM judges in both display orders, providing a resource for future studies of LLM-judge bias, heterogeneity, and human alignment.

1 Introduction

LLMs are increasingly used as scalable judges of model-generated responses, providing a practical complement to costly human evaluation (Zheng et al., 2023). However, their pairwise judgments can be sensitive to response display order, so observed preferences may partly reflect presentation instead of response quality (Wang et al., 2024a; Shi et al., 2025). Moreover, LLM and human evaluations can differ systematically (Chen et al., 2024; Gao et al., 2025), so removing position effects alone does not make an LLM judge human-aligned. Reliable human-targeted LLM evaluation must therefore address two distinct but coupled challenges: separating systematic position effects from underlying LLM preferences, and accounting for discrepancies between LLM and human preferences when human judgments are limited (Figure 1(a)). Figure 1(b) illustrates both effects empirically across LLM judges.

Existing work has studied these challenges largely along separate lines. Position-bias studies document order sensitivity (Wang et al., 2024a; Shi et al., 2025) and mitigate it via response swapping, verdict correction, or judge training (Wang et al., 2024a; Yang et al., 2025; Zhou et al., 2026; Yang et al., 2026), while human-supervised evaluation uses limited human labels to learn evaluation criteria, calibrate LLM-derived features, or improve human-centered evaluation (Liu et al., 2024; Kola et al., 2026; Boyeau et al., 2025; Xie et al., 2026). What remains less understood is how to connect stages in multi-judge evaluation: recover position-debiased LLM preferences, learn shared structure across judges, and calibrate it toward latent human preferences using limited human comparisons.

We introduce **DIAL** (Debiasing and human-preference Informed Alignment for LLM judges), a unified framework for position debiasing and human-preference calibration in multi-judge evaluation. DIAL separates judge-specific position effects from latent LLM preferences, learns shared structure across the recovered position-debiased preferences, and calibrates that structure toward a latent human-preference target using limited human comparisons. The resulting representation reduces unrestricted item-level human-preference estimation to calibration along learned low-dimensional preference directions. Adaptive weighting controls reliance on LLM evidence, interpolating between human-only and LLM-anchored estimation; Figure 1(c) summarizes the DIAL framework.

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Figure 1: Overview of DIAL framework. (a) Pairwise evaluation combines abundant, scalable LLM judgments with limited, higher-fidelity human judgments. (b) Empirical examples illustrate two distinct challenges: position bias and human-preference misalignment. (c) DIAL jointly models LLM and human comparisons to separate judge-specific position effects from latent LLM preferences, learn shared preference structure across judges, and adaptively calibrate toward human preference.

Contributions. Our contributions are threefold. (i) We characterize identification of judge-specific position effects and latent LLM preferences via a graph-cycle criterion, and identify the human target through the learned LLM preference space (Propositions 1 and 2). (ii) We characterize the first-order approximation-estimation tradeoff of LLM calibration (Theorem 3), derive fixed-weight inference and a local-misalignment shrinkage operator (Theorem 4 and Proposition D.6), and develop GACV for tuning the human-LLM weight using leave-one-human-out loss (Theorem 5). (iii) Across synthetic and three real benchmarks, DIAL remains robust to display imbalance and improves human-label efficiency under limited human supervision, including regimes where unrestricted human-only BTL estimation is unstable or nonfinite. Adaptive updating becomes especially valuable when the learned LLM representation is data-limited or imperfect. Our study also contributes over 410K judgments from 21 LLM judges across three benchmarks in both display orders.

2 Related Work

Position Bias and Paired-Comparison Models. LLM judges exhibit systematic biases including position, verbosity, and self-enhancement biases (Zheng et al., 2023): pairwise verdicts can change when the two responses are swapped (Wang et al., 2024a), and order sensitivity varies across judges, tasks, and comparison difficulty (Shi et al., 2025). Existing mitigation either corrects individual verdicts at evaluation time (Yang et al., 2025) or modifies the judge through training (Zhou et al., 2026; Yang et al., 2026). The Bradley-Terry-Luce (BTL) model (Bradley & Terry, 1952; Luce, 1959) can incorporate within-pair order effects (Davidson & Beaver, 1977; Augustin, 2004), treating display order as a comparison-varying covariate, so that identification becomes a property of the augmented comparison design, as in ranking models with comparison-varying covariates (Dong et al., 2026) and recent work on identifiability

of LLM-judge biases (Xu et al., 2026a). When position bias is not of concern, user-specific accuracy (Jin et al., 2020), judge-specific discrimination (Xu et al., 2026b) and consensus-disagreement structure (Yu et al., 2026) are studied for multiple judges.

Human Preference Aggregation and Calibration. Classical latent-variable approaches model annotator noise and reliability (Dawid & Skene, 1979; Chen et al., 2013), while empirical studies document systematic differences between human and LLM judges (Chen et al., 2024; Gao et al., 2025) and substantial uncertainty within human evaluation itself (Elangovan et al., 2025). Recent work uses limited human labels to learn evaluation criteria (Liu et al., 2024), to calibrate features extracted from LLM evaluations (Kola et al., 2026), or to combine abundant automatic labels with a smaller human sample when estimating aggregate evaluation functionals such as win rates, benchmark scores, or ranking metrics (Boyeau et al., 2025; Chen et al., 2026; Lee et al., 2026; Divekar & Majumder, 2026). DIAL instead targets latent human preferences and asks how a position-debiased multi-LLM-judge representation can improve estimation from limited human comparisons, separating order-effect estimation from human alignment and propagating uncertainty from both stages. See Section A for more detailed comparisons with related methods.

3 Position-debiased LLM preference representation

We write $s_H \in \mathbb{R}^N$ for pooled-human scores and $s_k \in \mathbb{R}^N$, $k = 1, \dots, K$, for the LLM judges' scores over the N items, and collect the LLM scores and order effects in $S = [s_1, \dots, s_K]^\top \in \mathbb{R}^{K \times N}$ and $b = (b_1, \dots, b_K)^\top \in \mathbb{R}^K$. Because BTL scores are invariant to common additive shifts, we fix centered representatives throughout.

Condition 1 (Score normalization). *For $k \in \{H, 1, \dots, K\}$, the latent scores satisfy $\mathbf{1}_N^\top s_k = 0$.*

3.1 Judge-specific order-effect BTL and position debiasing

For each unordered pair $\{i, j\}$, fix the canonical orientation $i < j$ and let $A \in \{-1, 1\}$ indicate whether item i is displayed first ($A = 1$) or second ($A = -1$). For judge k , let $\Omega_k = \{(i, j, a) : i < j, a \in \{-1, 1\}, n_{kij}^{(a)} > 0\}$ with $m_k = |\Omega_k|$, let $\mathcal{G}_k = ([N], \mathcal{E}_k)$ be the comparison graph in which $\{i, j\} \in \mathcal{E}_k$ whenever $(i, j, a) \in \Omega_k$ for at least one a , and let $Y_{kij}^{(a)}$ be the number of times judge k prefers item i among $n_{kij}^{(a)}$ comparisons under order $A = a$. For $(i, j, a) \in \Omega_k$, assume

$$Y_{kij}^{(a)} \sim \text{Binomial}\left(n_{kij}^{(a)}, p_{kij}^{(a)}\right), \quad \text{logit}\left(p_{kij}^{(a)}\right) = s_{ki} - s_{kj} + ab_k, \quad (1)$$

where $p_{kij}^{(a)}$ is the order-specific preference probability and $\sigma(x) = (1 + e^{-x})^{-1}$ denotes the logistic function. Conditional on the comparison design, the individual LLM comparison outcomes underlying these counts are mutually independent across judges, pairs, display orders, and repetitions. Here, $b_k > 0$ indicates preference for the first-displayed response and $b_k < 0$ preference for the second; exponentiating the score and order-effect terms recovers the classical within-pair order-effect BTL parameterization (Davidson & Beaver, 1977; Augustin, 2004), here applied judge-specifically to the response-order sensitivity documented for LLM evaluators (Wang et al., 2024a; Shi et al., 2025).

Fix an arbitrary enumeration $\Omega_k = \{(i_t, j_t, a_t)\}_{t=1}^{m_k}$. The item-difference design matrix $D_k \in \{-1, 0, 1\}^{m_k \times N}$ has rows $(D_k)_t = (e_{i_t} - e_{j_t})^\top$, and the order-indicator vector $z_k \in \{-1, 1\}^{m_k}$ has entries $(z_k)_t = a_t$.

Condition 2 (LLM comparison design). *For every $k = 1, \dots, K$, the augmented comparison design $[D_k, z_k]$ has rank N .*

Proposition 1 (Identification of latent scores and order effect). *Under (1) and Condition 1, for any judge k the parameter (s_k, b_k) is identifiable if and only if $\text{rank}([D_k, z_k]) = N$, equivalently if and only if $\mathcal{G}_k$ is connected and $z_k \notin \text{col}(D_k)$. Consequently, under Condition 2, (S, b) is identifiable.*

This rank characterization specializes augmented paired-comparison identification with comparison-varying covariates (Dong et al., 2026) to the observed order indicator; unlike unrestricted LLM-judge bias terms that can be confounded with item effects (Xu et al., 2026a), b_k is separable from s_k exactly when $z_k \notin \text{col}(D_k)$. Because $\text{col}(D_k)$ is the space of potential differences on the comparison multigraph, $z_k \notin \text{col}(D_k)$ holds if and only if some cycle has nonzero signed order sum (Corollary B.1), which is checkable in one pass; a pair shown under both orders is

the length-two cycle, and the criterion also covers designs in which every pair appears in one order only. Condition 2 is an identification condition on the comparison support and does not guarantee a finite empirical BTL estimator, which may fail under separation (Ford, 1957; Hunter, 2004); asymptotically, the same rank condition applies to the limiting support $\{(i, j, a) : \rho_{kij}^{(a)} > 0\}$ with within-judge limiting sampling proportions $\rho_{kij}^{(a)} := \lim n_{kij}^{(a)}/n_k$ and sample size $n_k := \sum_{(i', j', a') \in \Omega_k} n_{kij}^{(a')}$.

After separating the order effect, we define the position-debiased preference probability $p_{kij}^{\text{deb}} := \sigma(s_{ki} - s_{kj})$, so that judge k 's position-debiased ranking is read directly from s_k . This differs from the order-averaged probability $p_{kij}^{\text{avg}} := \{p_{kij}^{(1)} + p_{kij}^{(-1)}\}/2$, whose log-odds are a nonlinear odd function of $s_{ki} - s_{kj}$ whenever $b_k \neq 0$, so they are generally not representable by a single BTL score vector and BTL fitting to order-averaged data is generically misspecified (Section B.2). We therefore carry the recovered judge-specific s_k , rather than probability-averaged judgments, into the cross-judge representation and subsequent human calibration.

3.2 Structured representation across LLM judges

Following Yu et al. (2026), we adopt its consensus-disagreement decomposition for the position-debiased LLM score matrix.

Condition 3 (Structured LLM representation). *The score matrix S admits the representation*

$$S = \gamma\mu^\top + UV^\top, \quad (2)$$

where $\gamma \in \mathbb{R}^K$, $\mu \in \mathbb{R}^N$, $U \in \mathbb{R}^{K \times r}$, and $V \in \mathbb{R}^{N \times r}$ satisfy

- (i) $\mathbf{1}_N^\top \mu = 0$, $V^\top \mathbf{1}_N = 0$, $\mu^\top V = 0$, $N^{-1} \|\mu\|_2^2 = 1$ and $N^{-1} V^\top V = I_r$.
- (ii) $\mathbf{1}_K^\top \gamma > 0$, $\mathbf{1}_K^\top U = 0^\top$ and $\text{rank}(U) = r$.

Here μ is the common LLM preference direction, γ_k is judge k 's loading on it, and $UV^\top$ is rank- r heterogeneous variation whose item-side directions, the columns of V , are those along which the judges systematically disagree. These restrictions are compatible with Condition 1 and require $r \leq K - 1$ and $r + 1 \leq N - 1$. Define $W := [\mu, V] \in \mathbb{R}^{N \times (r+1)}$, so that (i) reads $W^\top \mathbf{1}_N = 0_{r+1}$ and $N^{-1} W^\top W = I_{r+1}$: fixing the scale on the item side rather than the judge side orthogonalizes the basis of $\text{col}(W)$, gives $\|Wc\|_2^2 = N\|c\|_2^2$, and confines the item side to a compact set. For human calibration in Section 4, let $M = \mu \in \mathbb{R}^{N \times 1}$ with $d = 1$, or $M = W \in \mathbb{R}^{N \times (r+1)}$ with $d = r + 1$, corresponding to consensus-only and full-space calibration, respectively.

Proposition 1 identifies (S, b) , and the constraints of Condition 3 then give $\mu = \sqrt{N} S^\top \mathbf{1}_K / \|S^\top \mathbf{1}_K\|_2$, $\gamma = N^{-1} S\mu$, $UV^\top = S - \gamma\mu^\top$, and $\text{col}(V) = \text{row}(UV^\top)$, so μ , γ , $UV^\top$, $\text{col}(V)$, and $\text{col}(W)$ are all identified. The factors U and V themselves are not, because $UV^\top = (UQ)(VQ)^\top$ for every orthogonal $Q \in \mathbb{R}^{r \times r}$; only the subspace $\text{col}(V)$ is invariant. Conditioning on $\text{col}(W)$, the shared structure reduces judge k 's design requirement from $\text{rank}([D_k, z_k]) = N$ to $\text{rank}([D_k W, z_k]) = r + 2$ (Proposition B.2). Since $\text{rank}(W) = r + 1$ and $\text{col}(W) = \text{span}(\mu) \oplus \text{col}(V)$, both calibration choices are defined by the position-debiased LLM representation.

4 Human preference alignment with limited supervision

4.1 Human calibration and identification

For pooled human judgments, recall that $s_H \in \mathbb{R}^N$ denotes the centered latent human preference score vector. For each pair $i < j$, let Y_{Hij} be the number of human comparisons preferring item i among n_{Hij} comparisons, define $n_H := \sum_{i < j} n_{Hij}$ and $\Omega_H := \{(i, j) : i < j, n_{Hij} > 0\}$, and let $\mathcal{G}_H$ denote the corresponding comparison graph. We assume

$$Y_{Hij} \sim \text{Binomial}(n_{Hij}, p_{Hij}), \quad \text{logit}(p_{Hij}) = s_{Hi} - s_{Hj}, \quad (i, j) \in \Omega_H. \quad (3)$$

Recent human-supervised LLM evaluation learns evaluation criteria or calibrates LLM-derived features from limited human labels (Liu et al., 2024; Kola et al., 2026); here the target is instead the latent pairwise human score s_H . Treating human annotations as noisy observations of this target, for a calibration basis M defined in Section 3.2, we impose the working restriction

$$s_H = M c_H, \quad c_H \in \mathbb{R}^d. \quad (4)$$

This reduces human estimation from $N - 1$ free score coordinates to d calibration coordinates. The restriction is an alignment assumption; $M c_H$ is basis-invariant, although for $M = W$ its coordinate c_H depends on the normalized basis for $\text{col}(V)$.

Enumerate $\Omega_H = \{(i_t, j_t)\}_{t=1}^{m_H}$, $m_H = |\Omega_H|$, and let $D_H \in \{-1, 0, 1\}^{m_H \times N}$ have rows $(e_{i_t} - e_{j_t})^\top$.

Condition 4 (Human comparison design). $\text{rank}(D_H M) = d$.

Proposition 2 (Identification under limited human comparisons). *Under model (1), (3), and (4), Conditions 1–4, c_H is identifiable relative to the chosen basis and $s_H = M c_H$ is identifiable and basis-invariant. The rank condition is equivalent to $\ker(D_H) \cap \text{col}(M) = \{0\}$; connectivity of $\mathcal{G}_H$ is sufficient but not necessary.*

Unlike methods combining automatic and human labels to estimate aggregate evaluation functionals (Boyeau et al., 2025; Chen et al., 2026; Lee et al., 2026; Divekar & Majumder, 2026), Proposition 2 targets the latent item-level preference through the learned calibration space $\text{col}(M)$. Condition 4 asks the human comparisons to distinguish only its d directions, so $m_H \geq d$ is necessary but connectivity or coverage of every item is not; Theorem C.5 extends the risk and inference results to incomplete population coverage.

Likelihoods. The normalized human loss for a centered score s is

$$\ell_H(s) := \frac{1}{n_H} \sum_{(i,j) \in \Omega_H} [n_{Hij} \log\{1 + \exp(s_i - s_j)\} - Y_{Hij}(s_i - s_j)]. \quad (5)$$

Equivalently $\ell_H(s) = n_H^{-1} \sum_{t=1}^{n_H} h_t(s)$, the average over individual comparisons, with $h_t(s) = \log\{1 + \exp(s_{I_t} - s_{J_t})\} - Z_t(s_{I_t} - s_{J_t})$ for the pair (I_t, J_t) and outcome $Z_t \in \{0, 1\}$. Let $n_L := \sum_{k=1}^K n_k$ be the total LLM sample size and define $\eta_{kij}^{(a)}(S, b) := S_{ki} - S_{kj} + ab_k$. The normalized LLM negative log-likelihood is

$$\ell_L(S, b) = \frac{1}{n_L} \sum_{k=1}^K \sum_{(i,j,a) \in \Omega_k} \left[n_{kij}^{(a)} \log\left\{1 + \exp\left(\eta_{kij}^{(a)}(S, b)\right)\right\} - Y_{kij}^{(a)} \eta_{kij}^{(a)}(S, b) \right]. \quad (6)$$

Calibration evaluates the same loss at $s = M c$. When a finite minimizer exists, define the unrestricted human-only estimator by

$$\hat{s}_0 \in \text{argmin}_{1_N^\top s=0} \ell_H(s). \quad (7)$$

4.2 Population value of LLM calibration

To characterize the tradeoff that the weighted estimator of Section 5.1 interpolates, we temporarily relax (4) and allow s_H to lie outside $\text{col}(M)$.

For the population analysis, we treat the n_H human comparisons as independent draws in which pair (i, j) is sampled with probability ρ_{ij} , with $\sum_{i < j} \rho_{ij} = 1$, and the outcome follows (3) conditional on the sampled pair. Define $\Omega_H^\infty := \{(i, j) : i < j, \rho_{ij} > 0\}$ and the corresponding comparison graph $\mathcal{G}_H^\infty$. The human comparison risk is $R_H(s) := \mathbb{E}\{h_t(s)\}$ under this sampling scheme. When $\mathcal{G}_H^\infty$ is connected, s_H minimizes R_H over centered score vectors.

Define the approximation error of the calibration space by

$$\Delta_M := \inf_{c \in \mathbb{R}^d} \{R_H(Mc) - R_H(s_H)\}. \quad (8)$$

For fixed sampling design ρ , Δ_M depends on M only through $\text{col}(M)$ and is nonnegative, vanishing if and only if $s_H \in \text{col}(M)$ when $\mathcal{G}_H^\infty$ is connected; Δ_M is design-dependent through ρ .

To isolate this tradeoff from estimation of the LLM representation, treat M as known at its population value and define the oracle anchored estimator $\tilde{s}_\infty := M \tilde{c}_\infty$, where $\tilde{c}_\infty \in \text{argmin}_{c \in \mathbb{R}^d} \ell_H(Mc)$. For expected-risk statements, we use the measurable extension of Lemma C.2, setting a fixed-space estimator to 0_N on its exponentially rare fallback event. The $(N - 1)/(2n_H)$ and $d/(2n_H)$ terms below are the classical first-order likelihood-risk complexity terms underlying AIC (Akaike, 1974).

Theorem 3 (Human-risk tradeoff from LLM calibration). *Under the sampling scheme above, let N and M be fixed, let $\mathcal{G}_H^\infty$ be connected, and assume the human probabilities are bounded away from 0 and 1 on Ω_H^∞ . If $s_H \in \text{col}(M)$, the first-order expected excess risks of $\hat{s}_0$ and $\tilde{s}_\infty$ are $(N-1)/(2n_H)$ and $d/(2n_H)$, respectively. More generally, let $s_H = s_H^{(n_H)}$ vary with n_H while keeping $\{\rho_{ij}\}$ fixed, and suppose its induced probabilities $p_{Hij}^{(n_H)} := \sigma(s_{Hi}^{(n_H)} - s_{Hj}^{(n_H)})$ lie in $[\varepsilon, 1-\varepsilon]$ on Ω_H^∞ for some fixed $\varepsilon > 0$, with $n_H \Delta_M \rightarrow \delta_M < \infty$. Then*

$$n_H \mathbb{E}\{R_H(\hat{s}_0) - R_H(s_H)\} \rightarrow \frac{N-1}{2}, \quad n_H \mathbb{E}\{R_H(\tilde{s}_\infty) - R_H(s_H)\} \rightarrow \delta_M + \frac{d}{2}. \quad (9)$$

Thus, anchoring has a smaller first-order risk exactly when $\delta_M < (N-1-d)/2$.

For $M = \mu$, $d = 1$ gives consensus-only calibration; for $M = W$, $d = r+1$ recovers full-space calibration. A fixed $\Delta_M > 0$ eventually dominates as n_H grows, motivating adaptive weighting. The above analysis treats M as fixed; the analysis also extends to misspecification of (4) and staged-oracle equivalence; see Theorem C.4 and Corollary C.6.

5 DIAL: Estimation and adaptive calibration

To utilize both the human and LLM comparisons, we combine the two likelihoods (5) and (6). For fixed rank $r \leq \min\{K-1, N-2\}$, let Θ_r denote the parameter space $\theta = (\gamma, \mu, U, V, b)$ satisfying Condition 3. For $\theta \in \Theta_r$, write $S_\theta := \gamma\mu^\top + UV^\top$ and $W_\theta := [\mu, V]$, and set $M_\theta := \mu$ for $M = \mu$ and $M_\theta := W_\theta$ for $M = W$.

5.1 Weighted likelihood estimator

Using the human loss in (5), define for $\lambda > 0$

$$Q_\lambda(s, S, b) := \ell_H(s) + \lambda \ell_L(S, b). \quad (10)$$

The factorized estimator is

$$(\text{DIAL-Ada}) \quad (\hat{c}_\lambda, \hat{\theta}_\lambda) \in \underset{c \in \mathbb{R}^d, \theta \in \Theta_r}{\operatorname{argmin}} Q_\lambda(M_\theta c, S_\theta, b), \quad \hat{s}_\lambda := M_{\hat{\theta}_\lambda} \hat{c}_\lambda. \quad (11)$$

Because the two likelihoods are normalized by n_H and n_L , λ controls their relative weight, with the ordinary joint likelihood corresponding to $\lambda = n_L/n_H$. Under exact calibration, that choice is efficient; choosing λ away from n_L/n_H trades exact-model efficiency for robustness when the calibration restriction is only approximate. We treat the score $\hat{s}_\lambda$ as the human-preference estimate.

We use the unrestricted human BTL estimator $\hat{s}_0$ in (7) as the $\lambda = 0$ endpoint. For $\lambda = \infty$, define the staged LLM-anchored endpoint by

$$(\text{DIAL-Anc}) \quad \hat{\theta}_\infty \in \underset{\theta \in \Theta_r}{\operatorname{argmin}} \ell_L(S_\theta, b), \quad \hat{c}_\infty \in \underset{c \in \mathbb{R}^d}{\operatorname{argmin}} \ell_H(M_{\hat{\theta}_\infty} c), \quad \hat{s}_\infty := M_{\hat{\theta}_\infty} \hat{c}_\infty. \quad (12)$$

Thus, $\lambda = 0$ gives human-only estimation, $\lambda = \infty$ gives LLM-first calibration, and finite λ jointly estimates the shared representation using both data sources. Relative to human-supervised calibration of LLM-derived features (Kola et al., 2026), finite λ additionally lets the human comparisons update the LLM representation itself, so $\hat{s}_\lambda$ is not restricted to a fixed estimated space. Under approximate calibration, this can bias the LLM parameters. For the fixed calibration choice $M \in \{\mu, W\}$, write $\mathcal{C}_\mu(S) := \text{span}(S^\top 1_K)$ and $\mathcal{C}_W(S) := \text{row}(S)$. For inference and tuning, we use Q_λ on the regular score space:

$$\mathcal{M}_r := \{(s, S, b) : S 1_N = 0_K, \text{rank}(S) = r+1, S^\top 1_K \neq 0_N, s \in \mathcal{C}_M(S), b \in \mathbb{R}^K\}, \quad (13)$$

Regular factorized and score-space fits have the same objective and calibrated score (Lemma D.1).

5.2 Inference at a fixed calibration weight

Fix $\lambda \in (0, \infty)$, N, K, r , and a calibration choice $M \in \{\mu, W\}$. Suppose the human comparisons are independent draws under the sampling scheme of Section 4.2, with connected population graph $\mathcal{G}_H^\infty$, independently of the LLM sample. Let $n_H, n_L \rightarrow \infty$ with $n_H/n_L \rightarrow \tau \in [0, \infty)$, and suppose $n_k/n_L \rightarrow \kappa_k > 0$ and $n_{kij}^{(a)}/n_k \rightarrow \rho_{kij}^{(a)}$, where each limiting within-judge support satisfies Condition 2. Write R_L for the limiting expected LLM loss as in (D.5) and use smooth local coordinates ζ on $\mathcal{M}_r$, with true coordinate ζ_0 . Define

$$\mathcal{I}_H := \nabla_{\zeta}^2 R_H\{s(\zeta)\}|_{\zeta=\zeta_0}, \quad \mathcal{I}_L := \nabla_{\zeta}^2 R_L\{S(\zeta), b(\zeta)\}|_{\zeta=\zeta_0},$$

together with $\mathcal{H}_\lambda := \mathcal{I}_H + \lambda \mathcal{I}_L$, $\mathcal{J}_{\lambda, \tau} := \mathcal{I}_H + \lambda^2 \tau \mathcal{I}_L$, and Jacobian $\dot{s}_0 := \frac{\partial s(\zeta_0)}{\partial \zeta^\top} \in \mathbb{R}^{N \times \dim \zeta}$. The covariance below has the standard sandwich form from likelihood and M-estimation theory (White, 1982); its two-sample form propagates both human and LLM sampling variation through $\mathcal{J}_{\lambda, \tau}$.

Theorem 4 (Fixed-weight inference for the calibrated preference). *Under model (1), (3), and (4), Conditions 1–3, and the sampling regime above,*

$$\sqrt{n_H}(\hat{s}_\lambda - s_H) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\lambda, s, \tau}), \quad \Sigma_{\lambda, s, \tau} := \dot{s}_0 \mathcal{H}_\lambda^{-1} \mathcal{J}_{\lambda, \tau} \mathcal{H}_\lambda^{-1} \dot{s}_0^\top. \quad (14)$$

The covariance is invariant to the smooth local coordinate used for $\mathcal{M}_r$. The empirical sandwich estimator $\hat{\Sigma}_{\lambda, s}$ defined in (D.8) satisfies $\hat{\Sigma}_{\lambda, s} \xrightarrow{P} \Sigma_{\lambda, s, \tau}$, so $n_H^{-1} \hat{\Sigma}_{\lambda, s}$ consistently estimates the first-order covariance of $\hat{s}_\lambda$. If $n_H/n_L \rightarrow 0$, the asymptotic linearization and sandwich consistency are uniform over $\lambda \in [\lambda_0, \infty)$ for every $\lambda_0 > 0$; Corollary D.5 also gives the $\lambda \rightarrow \infty$ staged limit.

For $\tau > 0$, both human and LLM sampling variation contribute through $\mathcal{J}_{\lambda, \tau}$; when $\tau = 0$, LLM sampling variation vanishes at the human $\sqrt{n_H}$ scale while its information remains through $\mathcal{H}_\lambda$. Since $1_N^\top s(\zeta) \equiv 0$, the limit law is supported on the centered subspace and $\Sigma_{\lambda, s, \tau} 1_N = 0_N$. For any centered contrast v with $v^\top \Sigma_{\lambda, s, \tau} v > 0$, including pairwise contrasts $v = e_i - e_j$ when nondegenerate, we studentize by $(v^\top \hat{\Sigma}_{\lambda, s} v / n_H)^{1/2}$. These intervals concern exact calibration at a fixed λ ; Theorem C.5 relaxes human-graph connectivity, and Proposition D.6 and Corollary D.7 give the local-misalignment limit.

5.3 Adaptive choice of the LLM weight

Let $\bar{\Lambda} \subset [0, \infty]$ be a fixed finite candidate set containing 0 and ∞ , and write $\Lambda := \bar{\Lambda} \setminus \{0, \infty\}$. Following the GACV principle of approximating leave-one-out likelihood loss without refitting every deletion (Xiang & Wahba, 1996) and related GCV-based tuning without sample splitting (Du et al., 2023), we select λ by approximating leave-one-human-comparison-out predictive loss while retaining all LLM comparisons in every candidate fit.

For each candidate fit, let $\hat{\zeta}_\lambda$ be its local-coordinate estimate and $q_\lambda(\zeta)$ its full-sample criterion, and define $\hat{H}_\lambda := \nabla_{\zeta}^2 q_\lambda(\hat{\zeta}_\lambda)$ and $\hat{J}_\lambda := n_H^{-1} \sum_t (g_{t, \lambda} - \bar{g}_\lambda)(g_{t, \lambda} - \bar{g}_\lambda)^\top$, where $g_{t, \lambda} := \nabla_{\zeta} h_t\{s(\zeta)\}|_{\zeta=\hat{\zeta}_\lambda}$ and $\bar{g}_\lambda := n_H^{-1} \sum_t g_{t, \lambda}$. Here, ζ denotes local coordinates on $\mathcal{M}_r$ for finite λ , on the centered score space at $\lambda = 0$, and on c with $M_{\hat{\theta}_\infty}$ fixed at $\lambda = \infty$. Define

$$\text{GACV}(\lambda) := \ell_H(\hat{s}_\lambda) + \frac{1}{n_H - 1} \text{tr} \left(\hat{H}_\lambda^{-1} \hat{J}_\lambda \right), \quad \hat{\lambda} \in \underset{\lambda \in \bar{\Lambda}}{\text{argmin}} \text{GACV}(\lambda). \quad (15)$$

For comparison, write $\text{CV}_{\text{loo}}(\lambda) := n_H^{-1} \sum_{t=1}^{n_H} h_t(\hat{s}_\lambda^{(-t)})$, where $\hat{s}_\lambda^{(-t)}$ is the leave-one-out (LOO) fit obtained after deleting human comparison t while retaining all LLM comparisons.

Theorem 5 (GACV tuning). *Suppose that N, K , and r are fixed, the calibration choice $M \in \{\mu, W\}$ is fixed, the human comparisons are independent draws under the sampling scheme of Theorem 4, independently of the LLM sample, and $\bar{\Lambda}$ is fixed and finite. Under Assumption A1 (verified under the correctly specified regime by Lemma D.8), as $n_H \rightarrow \infty$, conditionally on the LLM comparisons, $\text{GACV}(\lambda)$ is invariant to the chosen smooth local coordinates and*

$$\max_{\lambda \in \bar{\Lambda}} |\text{GACV}(\lambda) - \text{CV}_{\text{loo}}(\lambda)| = \mathcal{O}_{\mathbb{P}}(n_H^{-2}). \quad (16)$$

Here, GACV tunes DIAL-Ada (11) without a separate human validation sample or a rate restriction between n_H and n_L ; Corollary D.9 gives its near-LOO selection guarantee. The same criterion can select rank and weight jointly, while Proposition D.10 tests the calibration restriction and compares DIAL-Anc versus unrestricted human models by AIC.

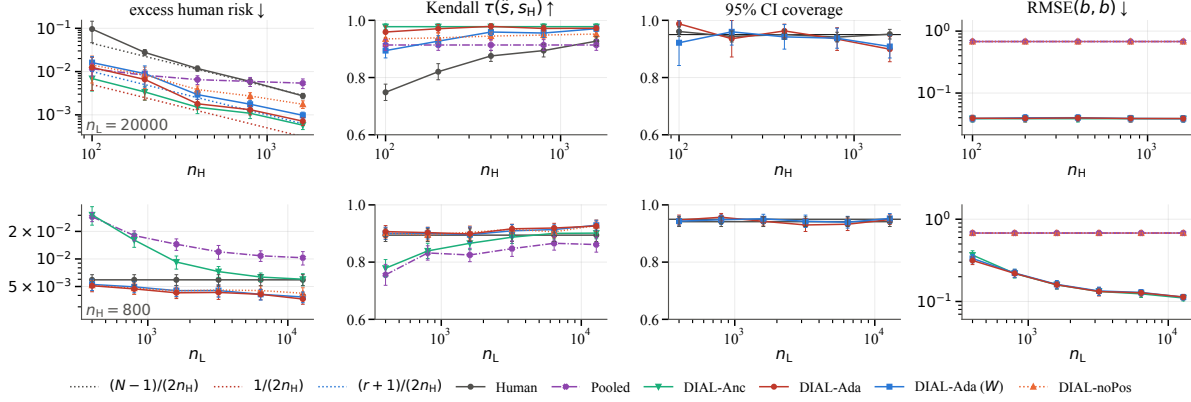


Figure 2: Synthetic evaluation with $N = 10$, $K = 4$, $r = 1$, $s_H = \mu$, and canonical-first display probability 0.75; curves show means ± 1.96 Monte Carlo standard errors over 50 replications. Columns report excess human risk, Kendall’s τ , 95% interval coverage, and RMSE($\hat{b}, b$). Top: n_H varies at $n_L = 20,000$ under correct specification; dotted lines are the first-order risk benchmarks for Human, calibration within $\text{span}(\mu)$, and calibration within $\text{col}(W)$. Bottom: n_L varies at $n_H = 800$ under judge-pair overdispersion that misspecifies the LLM likelihood while leaving the human calibration model correctly specified.

6 Experiments

We evaluate the two aspects of the proposed DIAL pipeline separately. Synthetic experiments first analyze recovery of order effects and position-debiased LLM preferences, and human-preference calibration under limited supervision and LLM misspecification. Real-data experiments study the same mechanisms against held-out human preferences.

Compared methods. We compare (i) Human, the unrestricted centered human BTL fit ($\lambda = 0$); (ii) Pooled, a pooled LLM BTL fit that ignores judge identity and display order and is rescaled using the human comparisons; (iii) DIAL-Ada and DIAL-Anc, the adaptive estimator (11) tuned by GACV and anchored estimator (12) within the consensus space; (iv) DIAL-Ada (W), the adaptive estimator (11) within W -space; (v) DIAL-noPos, the order-agnostic ablation of DIAL-Ada; and (vi) AtC, a stage-matched AtC baseline aggregating the same human comparisons by BTL and isotonic-calibrating the position-debiased LLM consensus.

6.1 Synthetic experiments

We use $N = 10$, $K = 4$, and $r = 1$, with structured LLM scores from (1) and consensus-aligned target $s_H = \mu$. The canonical item is displayed first with probability 0.75. In Figure 2, the top row varies n_H at $n_L = 20,000$ under exact specification, while the bottom row varies n_L at $n_H = 800$ under judge-pair overdispersion that misspecifies the LLM likelihood. Section F gives the full design and additional larger-scale, misalignment, and heterogeneous-position robustness experiments (Figures F1–F3).

Limited human supervision. Under exact specification, Human and DIAL-Anc closely follow their corresponding first-order risk benchmarks, with anchoring substantially improving both risk and ranking when human supervision is scarce. DIAL-Ada remains close to the anchored endpoint, while DIAL-Ada (W) pays additional variance for estimating an unnecessary disagreement direction. The order-agnostic Pooled and DIAL-noPos deteriorate under the unbalanced display.

Adaptive weighting and position recovery. Under pair-level LLM misspecification, the anchored estimator plateaus, whereas DIAL-Ada adapts toward the human data and remains competitive with or better than both endpoints, closely tracking the oracle candidate. Its ranking accuracy is correspondingly more stable as the LLM budget varies. For order-aware fits, order-effect estimation improves steadily with additional LLM comparisons, while order-agnostic methods retain substantial error; post-selection cell-clustered intervals remain empirically near nominal coverage.

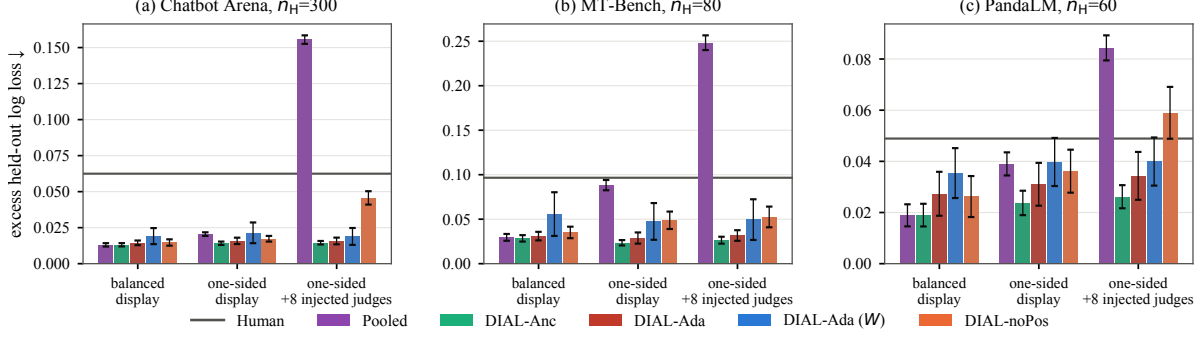


Figure 3: Real-data position robustness evaluation: excess held-out human log loss (zero is the test-pool BTL fit), averaged over up to 50 splits with ± 1.96 Monte Carlo standard errors. Panels show Chatbot Arena, MT-Bench, and PandaLM using the same five order-sensitive judges: Mistral-Small 3.2, Claude Haiku 4.5, Qwen3-30B without reasoning, GLM4-9B, and DeepSeek-R1-1.5B, under balanced, one-sided (5% swapped), and one-sided plus 8 injected pure-position judges.

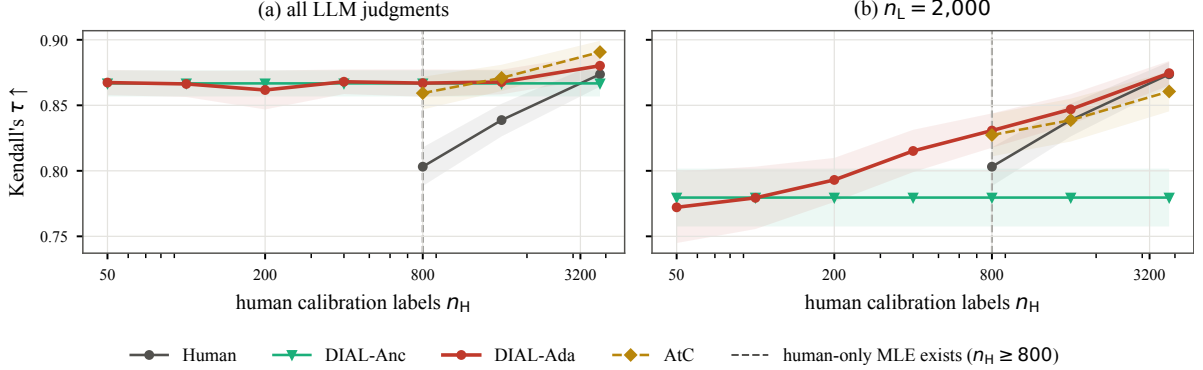


Figure 4: Human-label efficiency and adaptive calibration on Chatbot Arena: Kendall's τ with the held-out human ranking as n_H varies, using (a) all available judgments of the 21 LLM judges and (b) $n_L = 2,000$. Curves show means over up to 50 splits with ± 1.96 Monte Carlo standard errors; Human and AtC are shown only where the pooled-human BTL fit is finite in all splits, and the dashed vertical line marks the smallest such budget ($n_H = 800$).

6.2 Real-data experiments

We evaluate 21 LLM judges (18 open-weight via Ollama and 3 commercial API models) on Chatbot Arena (Chiang et al., 2024), MT-Bench (Zheng et al., 2023), and PandaLM (Wang et al., 2024b); after preprocessing, the benchmarks contain 20, 6, and 5 candidate models, respectively, with each LLM comparison queried in both display orders. We use disjoint human calibration and test sets and $r = 1$ throughout; position robustness evaluations report excess held-out human log loss, while calibration experiments report Kendall's τ against the held-out human ranking (Sections E.3 and G).

Order-aware DIAL remains stable under position perturbations. Under one-sided display, Pooled degrades markedly, and adding pure-position judges further degrades both Pooled and DIAL-noPos, whereas the order-aware DIAL fits remain stable across datasets (Figure 3). Judge-level order-effect diagnostics are reported in Figure G5(a).

Human efficiency and adaptive weighting. With all 21 LLM judges, DIAL-Anc and DIAL-Ada achieve strong held-out ranking agreement even at small human-label budgets, where the unrestricted Human fit is not finite in every split (Figure 4(a)). Where its rank-aggregation stage is finite, AtC gains 0.017–0.056 in τ over Human but stays within 0.013 of DIAL-Ada, measurably exceeding it only at the largest budget ($n_H = 3200$). With fewer LLM judgments, DIAL-Anc remains nearly flat as human supervision grows, whereas DIAL-Ada improves toward Human by updating the imperfect LLM consensus (Figure 4(b)). Appendix results show that DIAL-Ada remains more stable than fixed anchoring under anti-consensus corruption (Figure G4(c,f)), diagnose subgroup-specific failures of the

global calibration space (Figure G5(b)), and find similar human-label gains across smaller judge panels, with adaptive updating becoming more valuable as LLM data become scarce (Figures G7 and G8).

7 Discussion

DIAL treats position debiasing and human-preference alignment as distinct but complementary: removing display-order effects recovers cleaner LLM preferences, while limited human comparisons determine how those preferences should inform the human target. When the learned LLM preference structure is informative, it can substantially reduce human-label requirements; when it is imperfect, adaptive weighting shifts reliance toward the human data, as reflected in both theory and experiments. Key extensions include heterogeneous human preferences, richer position and tie models, post-selection inference, and cost-aware human comparison design.

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Appendix

This serves as an appendix to the main paper. Below, we provide an outline for the appendix along with a summary of the notation used in the main paper and the appendix.

Organization. The content of the appendix is organized as follows.

Appendix	Content
Section A	Detailed comparisons to related work.
Section B	B.1 Identification of position-debiased LLM scores and order effects (Proposition 1).
	B.2 Debiased versus order-averaged probabilities.
	B.3 Estimation of the position-debiased representation and judge diagnostics.
Section C	C.1 Identification and fixed-space risk preliminaries (Proposition 2).
	C.2 Proof of Theorem 3.
	C.3 Calibration under incomplete human comparison coverage.
	C.4 Additional results for human calibration (Corollary C.6).
Section D	D.1 Reduced score-space representation and regularity.
	D.2 Proof of Theorem 4.
	D.3 Proof of Theorem 5.
	D.4 Additional results for adaptive calibration (Corollary D.9 and Proposition D.10).
Section E	E.1 DIAL estimation procedure and adaptive GACV selection.
	E.2 Variance estimation for the DIAL intervals (cluster-robust HAC sandwich).
	E.3 Hyperparameter selection (rank r , weight λ).
Section F	F.1 Simulation design, estimation, and metrics.
	F.2 Larger-scale, misaligned-target, and heterogeneous-position simulations.
Section G	G.1 Datasets and collection.
	G.2 Protocol and estimators.
	G.3 Position robustness evaluations and diagnostics.
	G.4 Human-label efficiency and adaptive calibration.

Notation. For $d \geq 1$, let $[d] := \{1, \dots, d\}$, let e_j denote the j th standard basis vector in $\mathbb{R}^d$, and let 0_d and 1_d denote the zero and all-ones vectors; dimension subscripts are omitted when clear. For a set $\mathcal{S}$, $|\mathcal{S}|$ denotes its cardinality and $\mathbb{1}_{\{\cdot\}}$ the indicator function. For a matrix A , $\text{col}(A)$, $\text{row}(A)$, and $\text{rank}(A)$ denote its column space, row space, and rank, while $\|A\|_F$ and $\|A\|_{\text{op}}$ denote its Frobenius and operator norms. We write $\sigma(x) := (1 + e^{-x})^{-1}$ and $\text{logit}(p) := \log\{p/(1 - p)\}$ for the logistic function and its inverse. We use $o, \mathcal{O}$ and $o_{\mathbb{P}}, \mathcal{O}_{\mathbb{P}}$ for deterministic and stochastic order notation, and $\xrightarrow{p}$ and $\xrightarrow{d}$ for convergence in probability and distribution. Finally, $\text{KL}(p \parallel q)$ denotes the Kullback-Leibler divergence from p to q .

A Detailed Comparison with Closely Related Methods

This section separates DIAL’s contribution from established components: order-effect paired-comparison modeling and consensus-disagreement structure are prior ingredients, while DIAL links the resulting position-debiased multi-judge representation to limited-supervision human calibration and adaptive weighting.

Position bias and its identification. The closest work on position bias is Wang et al. (2024a), which shows that pairwise LLM judgments can change substantially under response-order permutations and develops calibration-based mitigation. Our position model instead treats display order as a comparison-varying covariate in a paired-comparison likelihood and separates the latent score contrast $s_{ki} - s_{kj}$ from the nuisance order effect ab_k . The resulting position-debiased preference is therefore defined by s_k , not by averaging judgments across display orders. As shown in Section B.2, probability-scale order averaging retains dependence on b_k and generally differs from the position-debiased probability, though the two induce the same pairwise ordering under our model. Balanced Position Calibration (Wang et al., 2024a) instead queries both display orders and aggregates the resulting evaluations. Under balanced querying, DIAL-noPos provides our closest order-agnostic comparison using both display orders, although its BTL aggregation differs from BPC’s score averaging; BPC itself requires the swapped judgment and is unavailable in the one-sided stress designs.

The identifiability results of Xu et al. (2026a) are relevant because they characterize when bias terms can or cannot be separated from unrestricted item effects. Their negative results apply when the bias contribution is confounded with item quality, and they develop additional comparison designs that supply the missing identifying information. DIAL instead treats the order indicator as a comparison-varying design covariate and identifies the order effect when its design column is linearly independent of the item-difference design, as formalized by Condition 2.

Structured preferences across LLM judges. Judge heterogeneity has also been modeled independently of position bias. Jin et al. (2020) model user-specific accuracy levels, while Xu et al. (2026b) introduce judge-specific discrimination parameters around a common latent ranking, capturing heterogeneity in how strongly judges respond to the shared ranking signal. Most directly, Yu et al. (2026) proposes the consensus-plus-heterogeneity decomposition $S = \gamma\mu^\top + UV^\top$, separating a consensus preference direction, judge-specific sensitivity to that direction, and structured residual disagreement. DIAL adopts this structured representation for the *position-debiased* LLM score matrix. After estimating and removing order effects, DIAL calibrates within either $\text{span}(\mu)$ or the full item-side space $\text{col}(W) = \text{span}(\mu) \oplus \text{col}(V)$. Accordingly, HJA targets interpretable decomposition and uncertainty quantification of heterogeneous judge preferences, whereas DIAL carries the learned representation into a separate human-alignment problem with population calibrated target Mc_M^* , where $c_M^* \in \text{argmin}_c R_H(Mc)$.

Human-supervised calibration and statistical evaluation. Several methods use human judgments to improve LLM-based evaluation: HD-Eval (Liu et al., 2024) learns human-aligned evaluation criteria and aggregation rules, while SAJA (Kola et al., 2026) uses a small human-labeled sample to calibrate features extracted from LLM evaluations toward human-aligned scores. In contrast, DIAL formulates the human target as a latent score and asks how limited pairwise human comparisons identify and estimate that score through a preference representation learned from LLM judges.

Aggregated evaluation functional estimation. A complementary statistical literature uses abundant automatic labels together with a smaller human reference sample to estimate evaluation quantities more efficiently. Boyeau et al. (2025), Chen et al. (2026), and Lee et al. (2026) study statistically principled estimation and uncertainty quantification for targets such as aggregate performance measures, benchmark scores, pairwise win rates, or evaluation accuracy, while Divekar & Majumder (2026) considers ranking metrics such as Precision@ K . These methods primarily improve the estimation of prespecified evaluation functionals. DIAL instead estimates the latent item-level preference s_H , or under approximation its calibrated counterpart Mc_M^* , from which pairwise probabilities and rankings are induced. As shown in Theorem 3, restricting human estimation to the learned LLM preference space trades approximation error against the estimation error.

Finally, AtC (Xie et al., 2026) first aggregates human comparisons into a consensus ordering and then isotonic-calibrates predictive scores to respect that ordering. It is the closest stage-matched alternative for our human-calibration

stage, but takes a predictive score vector and a human-derived ordering as inputs, instead of jointly estimating position effects and multi-judge preference structure. In Section 6.2, we therefore instantiate AtC using pooled BTL rank aggregation and the same LLM-only position-debiased consensus score as DIAL. DIAL instead learns the position-debiased LLM preference space $\text{col}(W)$ and calibrates a chosen $\text{col}(M) \subseteq \text{col}(W)$ from limited human comparisons. The weighted estimator in Section 5 further allows the amount of LLM anchoring to vary with human-LLM agreement, interpolating between unrestricted human-only estimation and staged LLM-anchored calibration.

B Supporting Results for Position Debiasing

B.1 Proof of Proposition 1

Proof of Proposition 1. Fix k and let $\delta := s_k - \tilde{s}_k$ and $c := b_k - \tilde{b}_k$ for two centered parameter values inducing the same comparison probabilities. Since the logistic link is one-to-one,

$$[D_k, z_k] \begin{pmatrix} \delta \\ c \end{pmatrix} = 0, \quad \mathbf{1}^\top \delta = 0.$$

Because $D_k \mathbf{1}_N = 0$, $(\mathbf{1}_N^\top, 0)^\top$ always belongs to $\ker([D_k, z_k])$. Hence centered identification holds if and only if this is the entire null space, equivalently $\text{rank}([D_k, z_k]) = N$; if the rank is smaller, centering the first block of any additional null vector gives a nonzero centered perturbation with the same comparison probabilities. Finally, $\text{rank}([D_k, z_k]) = N$ if and only if $\text{rank}(D_k) = N - 1$ and $z_k \notin \text{col}(D_k)$. For an incidence matrix, $\text{rank}(D_k) = N - 1$ is equivalent to connectivity of $\mathcal{G}_k$, which proves the equivalence. If a pair is observed under both orders, its two rows in D_k coincide while the corresponding entries of z_k are opposite, so $z_k \notin \text{col}(D_k)$. Applying the argument judge-wise proves identification of (S, b) under Condition 2. $\square$

Corollary B.1 (Cycle criterion for the order effect). *Suppose $\mathcal{G}_k$ is connected. Then b_k is identifiable if and only if some cycle of the comparison multigraph $\{(i_t, j_t)\}_{t=1}^{m_k}$, traversed with signs, has nonzero signed sum of the order labels a_t .*

Proof of Corollary B.1. By Proposition 1 it suffices to show that $z_k \in \text{col}(D_k)$ if and only if every signed cycle sum vanishes. An element of $\text{col}(D_k)$ has the form $(x_{i_t} - x_{j_t})_{t=1}^{m_k}$ for some $x \in \mathbb{R}^N$, that is, an edge function that is the difference of a potential, and a function on the oriented edges of a graph is a potential difference if and only if its signed sum around every cycle is zero. A pair observed under both orders contributes a cycle of length two with signed sum 2, which recovers the sufficient condition stated after Proposition 1. $\square$

Proposition B.2 (Identification under the low-rank structure). *Suppose Condition 3 holds. If $\text{col}(W)$ is known, then writing $s_k = W\alpha_k$ the parameter (α_k, b_k) of judge k is identifiable if and only if $\text{rank}([D_k W, z_k]) = r + 2$. If $\text{col}(W)$ is unknown, a necessary condition for identifiability of (S, b) is the parameter count $\sum_{k=1}^K m_k \geq (r + 1)(K + N - r - 2) + K$, and it is sufficient that some subset $\mathcal{K}_0$ of judges satisfies $\text{rank}([D_k, z_k]) = N$ for every $k \in \mathcal{K}_0$ together with $\text{rank}(S_{\mathcal{K}_0}) = r + 1$, in which case each remaining judge needs only $\text{rank}([D_k W, z_k]) = r + 2$.*

Proof of Proposition B.2. With $\text{col}(W)$ known the linear predictor is $D_k W\alpha_k + z_k b_k$, which determines (α_k, b_k) if and only if $[D_k W, z_k]$ has full column rank $r + 2$; no centering constraint is required because $\mathbf{1}_N^\top W = \mathbf{0}_{r+1}$. For unknown $\text{col}(W)$, removing the calibrated score from (D.4) leaves $\dim\{(S, b)\} = (r + 1)(K + N - r - 2) + K$, and identification requires at least as many comparison cells. For sufficiency, Proposition 1 identifies s_k for every $k \in \mathcal{K}_0$, the rank condition gives $\text{col}(W) = \text{row}(S_{\mathcal{K}_0})$, and the first part applies to each remaining judge. $\square$

A judge with too few comparisons to identify its own score vector can still be identified through the shared space, so the low-rankness in Condition 3 contributes to both identification and efficiency.

B.2 Debaised versus Order-Averaged Probabilities

Throughout this section, fix a judge k and suppress it from the notation: write $s_i, b, \mathcal{G}$, and $p_{ij}^{(a)}$ for $s_{ki}, b_k, \mathcal{G}_k$, and $p_{kij}^{(a)}$, and let $\Delta_{ij} := s_i - s_j$. We distinguish two mathematically related but conceptually different quantities.

- (i) **Position-debiased latent preference**, which defines the comparison probability by removing the order term from the fitted linear predictor and retaining only the latent score difference:

$$p_{ij}^{\text{deb}} := \sigma(\Delta_{ij}) = \sigma(s_i - s_j).$$

s represents the latent preference component separated from the nuisance order effect, so that the position-debiased ranking of the judge is obtained directly from s , $i \succ j$ iff $s_i > s_j$ iff $p_{ij}^{\text{deb}} > \frac{1}{2}$.

(ii) **Order-averaged protocol probability**, which averages the two order-specific probabilities:

$$p_{ij}^{\text{avg}} := \frac{1}{2} \left\{ p_{ij}^{(1)} + p_{ij}^{(-1)} \right\} = \frac{1}{2} \left\{ \sigma(\Delta_{ij} + b) + \sigma(\Delta_{ij} - b) \right\}.$$

Unlike p_{ij}^{deb} , it continues to depend on the magnitude of the judge's order effect.

DIAL uses (1), equivalently p_{ij}^{deb} , as the position-debiased latent preference of the judge, instead of p_{ij}^{avg} . Proposition B.3 characterizes the connection between the two quantities.

Proposition B.3 (Debiased versus order-averaged probabilities). *For a fixed pair $i < j$, the following hold:*

(i) *The position-debiased log-odds equal the average of the two order-specific log-odds:*

$$\text{logit}(p_{ij}^{\text{deb}}) = \Delta_{ij} = \frac{1}{2} \left\{ \text{logit}(p_{ij}^{(1)}) + \text{logit}(p_{ij}^{(-1)}) \right\}, \quad b = \frac{1}{2} \left\{ \text{logit}(p_{ij}^{(1)}) - \text{logit}(p_{ij}^{(-1)}) \right\}.$$

Thus p_{ij}^{deb} averages the two fitted comparisons on the log-odds scale, whereas p_{ij}^{avg} averages them on the probability scale.

(ii) *For finite Δ_{ij} , if $b \neq 0$ and $\Delta_{ij} \neq 0$,*

$$\left| p_{ij}^{\text{avg}} - \frac{1}{2} \right| < \left| p_{ij}^{\text{deb}} - \frac{1}{2} \right|.$$

Thus, probability-scale order averaging attenuates preferences toward $1/2$ whenever a nonzero order effect is present.

(iii) *For every finite b , p_{ij}^{avg} is a strictly increasing function of Δ_{ij} and satisfies $p_{ij}^{\text{avg}} = \frac{1}{2} \iff \Delta_{ij} = 0$. Consequently,*

$$p_{ij}^{\text{avg}} > \frac{1}{2} \iff p_{ij}^{\text{deb}} > \frac{1}{2} \iff s_i > s_j.$$

Proof of Proposition B.3. Let $u = \tanh(\Delta_{ij}/2)$ and $t = \tanh(b/2)$. For (i), the claim follows immediately from $\text{logit}(p_{ij}^{(1)}) = \Delta_{ij} + b$ and $\text{logit}(p_{ij}^{(-1)}) = \Delta_{ij} - b$. For (ii), using $\sigma(x) = \{1 + \tanh(x/2)\}/2$ and the addition formula for $\tanh$ gives

$$p_{ij}^{\text{avg}} - \frac{1}{2} = \frac{u}{2} \frac{1 - t^2}{1 - u^2 t^2} = \left(p_{ij}^{\text{deb}} - \frac{1}{2} \right) \frac{1 - t^2}{1 - u^2 t^2}. \quad (\text{B.1})$$

For finite Δ_{ij} and nonzero b , we have $0 < t^2 < 1$ and $u^2 < 1$, so

$$0 < \frac{1 - t^2}{1 - u^2 t^2} < 1.$$

Finally, for (iii), $\frac{\partial p_{ij}^{\text{avg}}}{\partial \Delta} = \frac{1}{2} \{ \sigma'(\Delta + b) + \sigma'(\Delta - b) \} > 0$ and symmetry gives $p_{ij}^{\text{avg}} = 1/2$ at $\Delta_{ij} = 0$. $\square$

As shown in Proposition B.3, although p_{ij}^{avg} and p_{ij}^{deb} induce the same pairwise ordering under the model, they generally differ in probability magnitude and calibration. The attenuation factor $(1 - t^2)/(1 - u^2 t^2)$ in (B.1) depends on the pair through $u = \tanh(\Delta_{ij}/2)$, so the matrix of order-averaged log-odds need not inherit the rank- $(r + 1)$ structure that Condition 3 imposes on S . By contrast p_{ij}^{deb} has a clean reading: its odds $\exp(\Delta_{ij})$ are the geometric mean of the two order-specific odds $\exp(\Delta_{ij} + b)$ and $\exp(\Delta_{ij} - b)$. Although order averaging preserves pairwise ordering, its log-odds are generally not exact score differences, so fitting a BTL model to the averaged probabilities is generically misspecified.

Proposition B.4 (Order averaging is not BTL). *If $b \neq 0$ and $\mathcal{G}$ contains a triangle, then for s outside a Lebesgue-null set there is no $\tilde{s} \in \mathbb{R}^N$ with $\text{logit}(p_{ij}^{\text{avg}}) = \tilde{s}_i - \tilde{s}_j$ on every compared pair.*

Proof of Proposition B.4. Let

$$g_b(\Delta) := \text{logit} \left[\frac{1}{2} \{ \sigma(\Delta + b) + \sigma(\Delta - b) \} \right].$$

By Proposition B.3(iii) and symmetry, g_b is odd, strictly increasing, and real analytic. For $b \neq 0$,

$$g'_b(0) = 4\sigma(b)\{1 - \sigma(b)\} < 1, \quad \lim_{\Delta \rightarrow \infty} g'_b(\Delta) = 1,$$

so g_b is nonlinear. By Corollary B.1, the edge values $g_b(\Delta_{ij})$ are score differences only if their signed sum vanishes on every cycle. On a triangle, this requires

$$F(x, y) := g_b(x + y) - g_b(x) - g_b(y) = 0.$$

Since F is real analytic and is not identically zero because g_b is nonlinear, its zero set has Lebesgue measure zero. $\square$

Proposition B.4 is an exact representability result and does not imply that the best BTL approximation must have large error. Indeed, for small b ,

$$g_b(\Delta) = \Delta - \frac{b^2}{2} \tanh(\Delta/2) + O(b^4),$$

while locally in Δ , $g_b(\Delta) = \text{sech}^2(b/2)\Delta + \mathcal{O}(\Delta^3)$. Thus, balanced order averaging can behave approximately as a rescaling of the latent score over the range of contrasts encountered in a dataset, even though no single score vector reproduces the averaged probabilities exactly.

B.3 Estimation of the Position-Debiased Representation

The LLM-only endpoint $\hat{\theta}_\infty$ of (12) is a maximum-likelihood estimator in the structured model of Condition 3. Its limit law supplies both the stability used by Corollary C.6 and the judge diagnostics for the calibrated preference. In particular, Proposition B.5 yields Wald statistics for $H_0 : b_k = 0$, testing whether judge k shows no order effect, and for $H_0 : b_k = b_{k'}$, testing whether two judges share one, together with intervals for the consensus loadings γ_k ; all are computed from the empirical inverse information at $\hat{\xi}_\infty$ at the $\sqrt{n_L}$ scale. Both hypotheses are basis-invariant: b is identified by Proposition 1, and $\gamma = N^{-1}S\mu$ is identified by the normalization in Condition 3. The argument parallels the fixed-rank likelihood analysis of Yu et al. (2026, Theorem 3), with the judge-wise comparison design augmented by the order indicator.

Proposition B.5 (Stage-one estimation). *Suppose that N , K , and r are fixed, that the LLM comparison model (1) and Condition 3 hold, that $n_k/n_L \rightarrow \kappa_k > 0$ and $n_{kij}^{(a)}/n_k \rightarrow \rho_{kij}^{(a)}$ with each limiting within-judge support satisfying Condition 2, and that every limiting cell probability lies in $(0, 1)$. Then, with probability tending to one, $\hat{\theta}_\infty$ is a regular minimizer of ℓ_L , and in any smooth coordinate ξ for the regular structured model*

$$\sqrt{n_L} (\hat{\xi}_\infty - \xi_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_\xi^{-1}), \quad \mathcal{I}_\xi := \nabla_\xi^2 R_L \{S(\xi), b(\xi)\} \big|_{\xi=\xi_0},$$

with $\mathcal{I}_\xi \succ 0$ by Condition 2. Consequently, $\hat{b}_\infty$, $\hat{\gamma}_\infty$, and $\hat{S}_\infty$ are $\sqrt{n_L}$ -consistent and asymptotically normal, and the estimated item-side space satisfies

$$\left\| P_{\text{col}(\hat{W}_\infty)} - P_{\text{col}(W)} \right\|_F = \mathcal{O}_{\mathbb{P}}(n_L^{-1/2}), \quad (\text{B.2})$$

where $P_{\text{col}(\cdot)}$ denotes orthogonal projection onto the indicated space.

Proof of Proposition B.5. By Proposition 1 and Condition 2, equality of the limiting LLM linear predictors implies equality of the centered parameter (S, b) . Hence, the limiting risk R_L is uniquely minimized at the truth, and in any smooth local coordinate ξ on the regular structured model, its Fisher information $\mathcal{I}_\xi$ is positive definite. The LLM-side sublevel-compactness argument in the proof of Lemma D.4, with the human component omitted, gives with probability tending to one a finite regular minimizer satisfying $\hat{\xi}_\infty \xrightarrow{p} \xi_0$. The Lindeberg-Feller score CLT, information identity, and Taylor expansion of the likelihood equation then give

$$\sqrt{n_L} (\hat{\xi}_\infty - \xi_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_\xi^{-1}).$$

The asymptotic normality of $\hat{b}_\infty$, $\hat{\gamma}_\infty$, and $\hat{S}_\infty$ follows by the delta method. Finally, Condition 3 gives $\text{col}(W) = \text{row}(S)$, and the map $S \mapsto P_{\text{row}(S)}$ is smooth on the rank- $(r + 1)$ stratum, so another application of the delta method yields (B.2). $\square$

C Supporting Theory for Human Preference Calibration

This section provides proofs and supporting results for human-preference calibration.

C.1 Identification and Fixed-Space Risk Preliminaries

Proof of Proposition 2. Under Conditions 1 and 2, Proposition 1 identifies (S, b) , and Condition 3 identifies the calibration space $\text{col}(M)$ with $\text{rank}(M) = d$ and $1_N^\top M = 0$. If c_H and $\tilde{c}_H$ induce the same human probabilities, then

$$D_H M(c_H - \tilde{c}_H) = 0,$$

so Condition 4 gives $c_H = \tilde{c}_H$. Conversely, failure of Condition 4 gives a nonzero coordinate perturbation with the same linear predictor. Since M has full column rank,

$$\text{rank}(D_H M) = d \iff \ker(D_H) \cap \text{col}(M) = \{0\}.$$

If $\mathcal{G}_H$ is connected, $\ker(D_H) = \text{span}(1_N)$, so the condition follows from $1_N^\top M = 0$. Finally, $M c_H$ is unchanged by the admissible basis transformation when $M = W$, while $M = \mu$ has no factor-basis ambiguity. $\square$

For any centered score vector $s \in \mathbb{R}^N$, define the population human negative log-likelihood

$$R_H(s) := \sum_{(i,j) \in \Omega_H^\infty} \rho_{ij} [\log\{1 + \exp(s_i - s_j)\} - p_{Hij}(s_i - s_j)]. \quad (\text{C.1})$$

For comparison with the full LLM preference space, define

$$\Delta_\mu := \inf_{\alpha \in \mathbb{R}} \{R_H(\alpha\mu) - R_H(s_H)\}, \quad (\text{C.2})$$

and let Δ_W be (8) at $M = W$; write δ_μ, δ_W for their local limits. For every centered $s \in \mathbb{R}^N$,

$$R_H(s) - R_H(s_H) = \sum_{(i,j) \in \Omega_H^\infty} \rho_{ij} \text{KL}\{\text{Bernoulli}(p_{Hij}) \parallel \text{Bernoulli}(\sigma(s_i - s_j))\}. \quad (\text{C.3})$$

Hence, if $\mathcal{G}_H^\infty$ is connected, $0 \leq \Delta_W \leq \Delta_\mu$, $\Delta_W = 0$ if and only if $s_H \in \text{col}(W)$, and $\Delta_\mu = 0$ if and only if $s_H \in \text{span}(\mu)$.

A risk expansion on a fixed score space. Let $B \in \mathbb{R}^{N \times q}$ have full column rank and satisfy $1_N^\top B = 0$. The corresponding centered score space is $\text{col}(B)$. We say that the population comparison support identifies $\text{col}(B)$ if

$$(e_i - e_j)^\top B u = 0 \text{ for all } (i, j) \in \Omega_H^\infty \implies u = 0. \quad (\text{C.4})$$

For $c \in \mathbb{R}^q$, define $R_B(c) := R_H(Bc)$. Under (C.4), the finite-support logistic risk is coercive and strictly convex, so it has a unique minimizer $c_B^* := \text{argmin}_{c \in \mathbb{R}^q} R_B(c)$. Let a generic human comparison sample $(I, J) \in \Omega_H^\infty$ with probability ρ_{ij} and then sample $Y \sim \text{Bernoulli}(p_{HIJ})$ conditional on (I, J) . Define

$$X_B := B^\top (e_I - e_J), \quad \pi_B := \sigma(X_B^\top c_B^*), \quad \psi_B := X_B(\pi_B - Y).$$

The population Hessian and score-variance matrices are

$$H_B := \mathbb{E}[\pi_B(1 - \pi_B)X_B X_B^\top], \quad J_B := \mathbb{E}[\psi_B \psi_B^\top]. \quad (\text{C.5})$$

The matrices H_B and J_B are the usual Hessian and score-variance quantities for maximum-likelihood estimation under possible model misspecification (White, 1982). Moreover, (C.4) implies $H_B \succ 0$.

The comparison Laplacian. A single matrix governs all of the population human quantities used below. Define the Fisher-weighted comparison Laplacian

$$L := \sum_{(i,j) \in \Omega_H^\infty} \rho_{ij} p_{Hij} (1 - p_{Hij}) (e_i - e_j)(e_i - e_j)^\top = \nabla^2 R_H(s_H), \quad (\text{C.6})$$

the Hessian of (C.1) at the truth. It is a weighted graph Laplacian for $\mathcal{G}_H^\infty$: $L \succeq 0$, $L1_N = 0_N$, and $\text{rank}(L) = N - 1$ if and only if $\mathcal{G}_H^\infty$ is connected. Write L^+ for the Moore-Penrose inverse of L , $L^{1/2}$ for its positive semidefinite square root and $L^{+/2} := (L^{1/2})^+$, $P := I_N - N^{-1}1_N1_N^\top$, and $P_A := A(A^\top A)^{-1}A^\top$ for A of full column rank. For a linear subspace $\mathcal{V} \subseteq \mathbb{R}^N$, define

$$\text{dist}_L^2\{x, \mathcal{V}\} := \inf_{v \in \mathcal{V}} (x - v)^\top L(x - v).$$

Lemma C.1 (Laplacian form of the fixed-space quantities). *Let $B \in \mathbb{R}^{N \times q}$ have full column rank with $1_N^\top B = 0$ and satisfy (C.4).*

- (i) *If the restricted model is correctly specified, then $H_B = J_B = B^\top LB$ in (C.5), and $\text{tr}(H_B^{-1}J_B) = q$.*
- (ii) *Suppose in addition that $\mathcal{G}_H^\infty$ is connected. If $\text{col}(B_H)$ is the whole centered score space, then $B_H(B_H^\top LB_H)^{-1}B_H^\top = L^+$, while*

$$B(B^\top LB)^{-1}B^\top = L^{+/2}P_{L^{1/2}B}L^{+/2} \preceq L^+,$$

with equality if and only if $q = N - 1$. The corresponding projection ranks are $\text{rank}(P) = N - 1$ and $\text{rank}(P_{L^{1/2}B}) = q$.

- (iii) *Suppose in addition that $\mathcal{G}_H^\infty$ is connected, and let $\epsilon_B := \inf_{c \in \mathbb{R}^q} \|Bc - s_H\|_2$. Along any local sequence with fixed ρ and comparison support, $p_{Hij} \in [\varepsilon, 1 - \varepsilon]$ for fixed $\varepsilon \in (0, 1/2)$, and $\epsilon_B \rightarrow 0$,*

$$\inf_{c \in \mathbb{R}^q} \{R_H(Bc) - R_H(s_H)\} = \frac{1}{2} \text{dist}_L^2\{s_H, \text{col}(B)\} + O(\epsilon_B^3).$$

Proof of Lemma C.1. For (i), correct specification gives $\pi_B = p_{HIJ}$ almost surely, so $H_B = \mathbb{E}[p_{HIJ}(1 - p_{HIJ})X_B X_B^\top] = B^\top LB$ by (C.6), while $\mathbb{E}[(\pi_B - Y)^2 | I, J] = p_{HIJ}(1 - p_{HIJ})$ gives $J_B = H_B$ and hence $\text{tr}(H_B^{-1}J_B) = q$.

For (ii), note first that $B(B^\top LB)^{-1}B^\top$ is unchanged when B is replaced by BA for invertible A , so we may take B_H with orthonormal columns spanning the centered space. Then $B_H B_H^\top = P$, and $L = PLP$ gives $L = B_H(B_H^\top LB_H)^{-1}B_H^\top$ with $B_H^\top LB_H$ invertible by connectivity; the matrix $B_H(B_H^\top LB_H)^{-1}B_H^\top$ then satisfies the four Moore-Penrose conditions for L , so it equals L^+ . Next, $P_{L^{1/2}B} = L^{1/2}B(B^\top LB)^{-1}B^\top L^{1/2}$, and $L^{+/2}L^{1/2} = P$ with $PB = B$, so $L^{+/2}P_{L^{1/2}B}L^{+/2} = B(B^\top LB)^{-1}B^\top$; the same computation with B replaced by B_H gives $L^+ = L^{+/2}P_{L^{1/2}B}L^{+/2}$. Since $\text{col}(L^{1/2}B) \subseteq \text{col}(P)$ we have $P - P_{L^{1/2}B} \succeq 0$, and conjugating by $L^{+/2}$ preserves this, with equality exactly when $\text{col}(L^{1/2}B) = \text{col}(P)$, that is when $q = N - 1$.

For (iii), connectivity and the uniform interior bound make L uniformly positive definite on the centered subspace. Let $\bar{s}_B$ minimize $(s - s_H)^\top L(s - s_H)$ over $s \in \text{col}(B)$. Norm equivalence gives $\|\bar{s}_B - s_H\|_2 = O(\epsilon_B)$. Since $\nabla R_H(s_H) = 0_N$, $\nabla^2 R_H(s_H) = L$, and the third derivatives are uniformly bounded,

$$R_H(\bar{s}_B) - R_H(s_H) = \frac{1}{2} \text{dist}_L^2\{s_H, \text{col}(B)\} + O(\epsilon_B^3).$$

The restricted risk minimizer has no larger risk, and local strong convexity gives distance $O(\epsilon_B)$ from s_H . Applying the same Taylor expansion there and using optimality of $\bar{s}_B$ for the quadratic term gives the reverse bound. $\square$

Taking $B = M$ in Lemma C.1 gives the calibration-space variance reduction and the corresponding local expansion for Δ_M .

For the following fixed-space proofs, write $\ell_{H,B}(c) := \ell_H(Bc)$, with all derivatives taken with respect to c . The estimation regularity that the risk expansion requires is not an assumption but a consequence of the finite design.

Lemma C.2 (Existence, consistency, and moment bounds on a fixed score space). *Suppose (C.4) holds for B , that Ω_H^∞ is finite with $\rho_{ij} > 0$, and that $p_{Hij} \in [\varepsilon, 1 - \varepsilon]$ for a fixed $\varepsilon \in (0, 1/2)$. Let $\mathcal{E}$ be the event that every cell of Ω_H^∞ is observed at least once with each outcome, and take $\hat{c}_B$ to be the minimizer of $\ell_{H,B}$ on $\mathcal{E}$ and $\hat{c}_B := 0$ off $\mathcal{E}$. Then there are constants $c, C > 0$, depending on p_H only through ε , such that for every n_H :*

- (i) $\Pr(\mathcal{E}^c) \leq Ce^{-cn_H}$, and on $\mathcal{E}$ the criterion $\ell_{H,B}$ has a unique finite minimizer, which satisfies $\|\hat{c}_B\|_2 \leq Cn_H$;
- (ii) $\Pr\{\sqrt{n_H}\|\hat{c}_B - c_B^*\|_2 > t\} \leq Ce^{-ct^2}$ for every $0 \leq t \leq c\sqrt{n_H}$;

(iii) consequently $\widehat{c}_B \xrightarrow{p} c_B^*$, and $n_H \|\widehat{c}_B - c_B^*\|_2^2$ is uniformly integrable.

Because the constants depend on p_H only through ε , all three statements hold uniformly over triangular arrays $\{p_{Hij}^{(n_H)}\}$ obeying the same bound.

Proof of Lemma C.2. Write $x_{ij} := B^\top(e_i - e_j)$. Call $\mathcal{A} \subseteq \Omega_H^\infty$ identifying if $x_{ij}^\top u = 0$ for all $(i, j) \in \mathcal{A}$ forces $u = 0$; for such an $\mathcal{A}$ the map $u \mapsto \max_{(i,j) \in \mathcal{A}} |x_{ij}^\top u|$ is a norm on $\mathbb{R}^q$, and we let $\kappa > 0$ be the smallest of its minima over the unit sphere, taken over the finitely many identifying subsets.

For (i), a cell observed with both outcomes contributes at least $\log(1+e^v) + \log(1+e^{-v}) \geq |v|$ to $n_H \ell_{H,B}(c)$, where $v := x_{ij}^\top c$, because $\log(1+e^v) \geq \max(0, v)$. On $\mathcal{E}$ this holds for every cell, so $n_H \ell_{H,B}(c) \geq \kappa \|c\|_2$, and the criterion is coercive; (C.4) makes it strictly convex, so the minimizer exists and is unique. Since $\ell_{H,B}(\widehat{c}_B) \leq \ell_{H,B}(0) = \log 2$, the same bound gives $\|\widehat{c}_B\|_2 \leq n_H \log 2 / \kappa$. For the probability, $n_{Hij} \sim \text{Binomial}(n_H, \rho_{ij})$ and, conditionally on n_{Hij} , the chance that a cell shows a single outcome is at most $2(1 - \varepsilon)^{n_{Hij}}$, so

$$\Pr(\mathcal{E}^c) \leq \sum_{(i,j) \in \Omega_H^\infty} \left\{ \Pr(n_{Hij} < \rho_{ij} n_H / 2) + 2(1 - \varepsilon)^{\rho_{ij} n_H / 2} \right\} \leq C e^{-c n_H},$$

since Ω_H^∞ is finite and each $\rho_{ij} > 0$.

For (ii), $\nabla \ell_{H,B}(c_B^*)$ is an average of n_H independent, mean-zero, uniformly bounded vectors, so Hoeffding's inequality gives $\Pr\{\sqrt{n_H} \|\nabla \ell_{H,B}(c_B^*)\|_2 > t\} \leq C e^{-ct^2}$. On a fixed ball $\|c - c_B^*\|_2 \leq \delta$ the logistic weights are bounded below, and the empirical cell frequencies concentrate, so $\nabla^2 \ell_{H,B} \succeq \kappa' I_q$ there for some $\kappa' > 0$ with probability at least $1 - C e^{-c n_H}$. Convexity then gives $\|\widehat{c}_B - c_B^*\|_2 \leq \kappa'^{-1} \|\nabla \ell_{H,B}(c_B^*)\|_2$ whenever the right-hand side is at most $\kappa' \delta$, which is the stated range of t .

For (iii), consistency follows from (ii). For uniform integrability, first note that c_B^* is uniformly bounded under the stated interior-probability condition. Indeed, for every $p \in [\varepsilon, 1 - \varepsilon]$ and $v \in \mathbb{R}$,

$$\log(1 + e^v) - pv \geq \varepsilon |v|.$$

Hence, writing $\rho_{\min} := \min_{(i,j) \in \Omega_H^\infty} \rho_{ij} > 0$ and using the definition of κ above,

$$\begin{aligned} R_B(c) &\geq \varepsilon \sum_{(i,j) \in \Omega_H^\infty} \rho_{ij} |x_{ij}^\top c| \\ &\geq \varepsilon \rho_{\min} \kappa \|c\|_2. \end{aligned}$$

Since $R_B(c_B^*) \leq R_B(0) = \log 2$, it follows that

$$\|c_B^*\|_2 \leq \frac{\log 2}{\varepsilon \rho_{\min} \kappa}.$$

Now set

$$Z_{n_H} := \sqrt{n_H} \|\widehat{c}_B - c_B^*\|_2, \quad X_{n_H} := Z_{n_H}^2 = n_H \|\widehat{c}_B - c_B^*\|_2^2.$$

We prove uniform integrability by showing that $\{X_{n_H}\}$ has uniformly bounded second moments. Choose a fixed $a > 0$ small enough that (ii) applies for every $0 \leq t \leq a\sqrt{n_H}$. Using the tail-integral representation,

$$\begin{aligned} \mathbb{E}[Z_{n_H}^4 \mathbb{1}_{\{Z_{n_H} \leq a\sqrt{n_H}\}}] &\leq 4 \int_0^{a\sqrt{n_H}} t^3 \Pr(Z_{n_H} > t) dt \\ &\leq C \int_0^\infty t^3 e^{-ct^2} dt \leq C. \end{aligned}$$

It remains to control the far tail. On $\mathcal{E}$, part (i) and the uniform bound on c_B^* give

$$Z_{n_H} \leq C n_H^{3/2}.$$

Hence, applying (ii) at $t = a\sqrt{n_H}$,

$$\begin{aligned} \mathbb{E}[Z_{n_H}^4 \mathbb{1}_{\{Z_{n_H} > a\sqrt{n_H}\}} \mathbb{1}_{\mathcal{E}}] &\leq C n_H^6 \Pr(Z_{n_H} > a\sqrt{n_H}) \\ &\leq C n_H^6 e^{-c n_H} \leq C. \end{aligned}$$

Off $\mathcal{E}$, the convention $\widehat{c}_B = 0$ and the uniform bound on c_B^* imply

$$Z_{n_H}^4 = n_H^2 \|c_B^*\|_2^4 \leq C n_H^2,$$

and therefore

$$\mathbb{E}[Z_{n_H}^4 \mathbf{1}_{\mathcal{E}^c}] \leq C n_H^2 e^{-c n_H} \leq C.$$

Consequently,

$$\sup_{n_H} \mathbb{E}[X_{n_H}^2] = \sup_{n_H} \mathbb{E}[Z_{n_H}^4] < \infty.$$

It follows that, for every $A > 0$,

$$\sup_{n_H} \mathbb{E}[X_{n_H} \mathbf{1}_{\{X_{n_H} > A\}}] \leq \frac{1}{A} \sup_{n_H} \mathbb{E}[X_{n_H}^2] \rightarrow 0 \quad (A \rightarrow \infty).$$

Thus $n_H \|\widehat{c}_B - c_B^*\|_2^2$ is uniformly integrable. □

The convention in Lemma C.2 modifies the empirical minimizer only on an event of probability at most Ce^{-cn_H} , so it affects neither the limit distribution nor the $o(n_H^{-1})$ risk expansion below.

Lemma C.3 (Risk expansion on a fixed score space). *Suppose that the conditions of Lemma C.2 hold. Then*

$$\sqrt{n_H}(\widehat{c}_B - c_B^*) \xrightarrow{d} \mathcal{N}(0, H_B^{-1} J_B H_B^{-1}),$$

and

$$\mathbb{E}[R_H(B\widehat{c}_B) - R_H(Bc_B^*)] = \frac{1}{2n_H} \text{tr}(H_B^{-1} J_B) + o(n_H^{-1}). \quad (\text{C.7})$$

If the restricted model is correctly specified, then $H_B = J_B$ and the leading term in (C.7) equals $q/(2n_H)$. The remainder is uniform over triangular arrays $\{p_{Hij}^{(n_H)}\}$ with $p_{Hij}^{(n_H)} \in [\varepsilon, 1 - \varepsilon]$ and $H_B \succeq \kappa I_q$ for fixed $\varepsilon, \kappa > 0$.

Proof of Lemma C.3. Recall the local shorthand $\ell_{H,B}(c) = \ell_H(Bc)$, differentiated with respect to c . At the population minimizer,

$$\mathbb{E}\{\nabla \ell_{H,B}(c_B^*)\} = \nabla R_B(c_B^*) = 0.$$

Because the pair support is finite and the outcomes are Bernoulli, the multivariate central limit theorem gives

$$\sqrt{n_H} \nabla \ell_{H,B}(c_B^*) \xrightarrow{d} \mathcal{N}(0, J_B).$$

Let $\mathcal{A}_{n_H}$ denote the event that a finite empirical minimizer exists. On $\mathcal{A}_{n_H}$, a Taylor expansion of the empirical score around c_B^* gives

$$0 = \nabla \ell_{H,B}(\widehat{c}_B) = \nabla \ell_{H,B}(c_B^*) + \{H_B + o_{\mathbb{P}}(1)\}(\widehat{c}_B - c_B^*),$$

where consistency and continuity of the logistic Hessian imply convergence of the empirical Hessian. Since $\Pr(\mathcal{A}_{n_H}) \rightarrow 1$,

$$\sqrt{n_H}(\widehat{c}_B - c_B^*) = -H_B^{-1} \sqrt{n_H} \nabla \ell_{H,B}(c_B^*) + o_{\mathbb{P}}(1),$$

which proves the stated asymptotic normality.

Since c_B^* minimizes R_B , $\nabla R_B(c_B^*) = 0$. Let $u_B := \widehat{c}_B - c_B^*$ and define

$$\overline{H}_B := 2 \int_0^1 (1-t) \nabla^2 R_B(c_B^* + t u_B) dt.$$

The integral form of Taylor's theorem gives

$$R_B(\widehat{c}_B) - R_B(c_B^*) = \frac{1}{2} u_B^\top \overline{H}_B u_B.$$

Because $\widehat{c}_B \xrightarrow{p} c_B^*$ and $\nabla^2 R_B$ is continuous,

$$\overline{H}_B \xrightarrow{p} H_B.$$

Moreover, the pair support is finite and $\sigma(x)\{1 - \sigma(x)\} \leq 1/4$, so

$$\sup_{c \in \mathbb{R}^q} \|\nabla^2 R_B(c)\|_{\text{op}} < \infty.$$

Hence, there exists some constant $C > 0$ such that

$$0 \leq n_{\text{H}} \{R_B(\hat{c}_B) - R_B(c_B^*)\} \leq C n_{\text{H}} \|\hat{c}_B - c_B^*\|_2^2.$$

By Lemma C.2(iii), the right-hand side is uniformly integrable. Together with the stated asymptotic normality and $\bar{H}_B \xrightarrow{p} H_B$, this yields

$$n_{\text{H}} \{R_B(\hat{c}_B) - R_B(c_B^*)\} \xrightarrow{d} \frac{1}{2} Z^\top H_B Z, \quad Z \sim \mathcal{N}(0, H_B^{-1} J_B H_B^{-1}),$$

and uniform integrability therefore gives

$$\begin{aligned} \lim_{n_{\text{H}} \rightarrow \infty} n_{\text{H}} \mathbb{E} [R_B(\hat{c}_B) - R_B(c_B^*)] &= \frac{1}{2} \mathbb{E} (Z^\top H_B Z) \\ &= \frac{1}{2} \text{tr}(H_B^{-1} J_B). \end{aligned}$$

This proves (C.7).

If the restricted model is correctly specified, then $\pi_B = p_{\text{H}IJ}$ almost surely. The information identity gives $J_B = H_B$, and therefore $\text{tr}(H_B^{-1} J_B) = q$. $\square$

C.2 Proof of Theorem 3

Throughout this subsection, fixed-space empirical minimizers use the measurable extension in Lemma C.2.

Theorem C.4 (Risk on a fixed score space). *Let N be fixed and let $\mathcal{V}$ be a fixed q -dimensional subspace of the centered score space on which the population comparison support is identifying in the sense of (C.4). Let $H_{\mathcal{V}}$ and $J_{\mathcal{V}}$ denote (C.5) in any basis of $\mathcal{V}$. Write $\tilde{s}_{\mathcal{V}}$ for the minimizer of ℓ_{H} over $\mathcal{V}$ and $\Delta_{\mathcal{V}} := \inf_{s \in \mathcal{V}} R_{\text{H}}(s) - R_{\text{H}}(s_{\text{H}})$. Under the sampling scheme of Section 4.2,*

$$\mathbb{E} [R_{\text{H}}(\tilde{s}_{\mathcal{V}}) - R_{\text{H}}(s_{\text{H}})] = \Delta_{\mathcal{V}} + \frac{1}{2n_{\text{H}}} \text{tr}(H_{\mathcal{V}}^{-1} J_{\mathcal{V}}) + o(n_{\text{H}}^{-1}). \quad (\text{C.8})$$

Under correct specification the estimation term is $q/(2n_{\text{H}}) + o(n_{\text{H}}^{-1})$. The whole centered score space gives $q = N - 1$, while $\mathcal{V} = \text{col}(M)$ gives $q = d$. If, along a local-mismatch sequence with fixed ρ , $p_{\text{H}ij}^{(n_{\text{H}})} \in [\varepsilon, 1 - \varepsilon]$ on $\Omega_{\text{H}}^{\infty}$ for fixed $\varepsilon \in (0, 1/2)$ and $n_{\text{H}} \Delta_{\mathcal{V}} \rightarrow \delta_{\mathcal{V}} < \infty$, then

$$n_{\text{H}} \mathbb{E} \{R_{\text{H}}(\tilde{s}_{\mathcal{V}}) - R_{\text{H}}(s_{\text{H}})\} \rightarrow \delta_{\mathcal{V}} + \frac{q}{2},$$

so anchoring to $\mathcal{V}$ improves first-order risk exactly when $\delta_{\mathcal{V}} < (N - 1 - q)/2$.

Proof of Theorems 3 and C.4. Let B be any basis of $\mathcal{V}$, so that $1_N^\top B = 0$ and B has full column rank q . Both $\tilde{s}_{\mathcal{V}} = B\hat{c}_B$ and $\inf_{s \in \mathcal{V}} R_{\text{H}}(s) = R_B(c_B^*)$ are invariant to replacing B by BA for invertible A . By the definition of $\Delta_{\mathcal{V}}$,

$$\mathbb{E} [R_{\text{H}}(\tilde{s}_{\mathcal{V}}) - R_{\text{H}}(s_{\text{H}})] = \Delta_{\mathcal{V}} + \mathbb{E} [R_B(\hat{c}_B) - R_B(c_B^*)],$$

and Lemma C.3 applied to B gives (C.8). If $s_{\text{H}} \in \mathcal{V}$, then $\Delta_{\mathcal{V}} = 0$ because s_{H} minimizes R_{H} globally, the restricted model is correctly specified, and $\text{tr}(H_{\mathcal{V}}^{-1} J_{\mathcal{V}}) = q$ by Lemma C.1(i).

For the unrestricted endpoint, take $B_{\text{H}} \in \mathbb{R}^{N \times (N-1)}$ spanning the centered score space; connectivity of $\mathcal{G}_{\text{H}}^{\infty}$ gives (C.4), correct specification, and $q = N - 1$. For the anchored endpoint, take $B = M$ and $q = d$; (C.4) is equivalent to $\text{rank}(D_{\text{H}}^{\infty} M) = d$, as relaxed further in Theorem C.5.

Local-mismatch regime. Let $\pi_{ij}^* := \sigma\{(e_i - e_j)^\top B c_B^*\}$. By the KL identity (C.3), which does not require connectivity,

$$\Delta_{\mathcal{V}} = \sum_{(i,j) \in \Omega_{\mathcal{H}}^\infty} \rho_{ij} \text{KL} \{ \text{Bernoulli}(p_{\mathcal{H}ij}) \parallel \text{Bernoulli}(\pi_{ij}^*) \} \geq 0.$$

Hence $n_{\mathcal{H}} \Delta_{\mathcal{V}} \rightarrow \delta_{\mathcal{V}} < \infty$ implies $\Delta_{\mathcal{V}} \rightarrow 0$ and therefore $\pi_{ij}^* - p_{\mathcal{H}ij} \rightarrow 0$ for every $(i, j) \in \Omega_{\mathcal{H}}^\infty$. Specializing (C.5) gives $J_{\mathcal{V}} - H_{\mathcal{V}} \rightarrow 0$, while the uniform interior bound keeps $H_{\mathcal{V}}$ uniformly positive definite, so $\text{tr}(H_{\mathcal{V}}^{-1} J_{\mathcal{V}}) \rightarrow q$. The remainder in (C.8) is uniform along the sequence by the triangular-array clause of Lemma C.3, so multiplying by $n_{\mathcal{H}}$ gives

$$n_{\mathcal{H}} \mathbb{E} [R_{\mathcal{H}}(\tilde{s}_{\mathcal{V}}) - R_{\mathcal{H}}(s_{\mathcal{H}})] \longrightarrow \delta_{\mathcal{V}} + \frac{q}{2},$$

and the same argument with $\mathcal{V}$ the whole centered score space gives the limit $(N - 1)/2$. The former is smaller if and only if $\delta_{\mathcal{V}} < (N - 1 - q)/2$. Taking $\mathcal{V} = \text{col}(M)$ and $q = d$ gives Theorem 3. $\square$

C.3 Calibration under Incomplete Human Comparison Coverage

Under (4), connectivity of $\mathcal{G}_{\mathcal{H}}^\infty$ can be replaced by identification of the d directions in $\text{col}(M)$. Enumerate $\Omega_{\mathcal{H}}^\infty = \{(i_t, j_t)\}_{t=1}^{m_{\mathcal{H}}^\infty}$, let $D_{\mathcal{H}}^\infty$ have rows $(e_{i_t} - e_{j_t})^\top$, and write $C_{\mathcal{H}}$ for the number of connected components of $\mathcal{G}_{\mathcal{H}}^\infty$, including isolated items, so $q_{\mathcal{H}} := \text{rank}(D_{\mathcal{H}}^\infty) = N - C_{\mathcal{H}}$.

Theorem C.5 (Calibration with incomplete human coverage). *Suppose (4) holds and $\text{rank}(D_{\mathcal{H}}^\infty M) = d$. Then:*

- (i) $c_{\mathcal{H}}$ and the basis-invariant score $s_{\mathcal{H}} = M c_{\mathcal{H}}$ are identifiable from the population human probabilities, whereas an unrestricted centered score is not identifiable when $C_{\mathcal{H}} > 1$.
- (ii) Treating M as known,

$$\mathbb{E}[R_{\mathcal{H}}(\tilde{s}_\infty) - R_{\mathcal{H}}(s_{\mathcal{H}})] = \frac{d}{2n_{\mathcal{H}}} + o(n_{\mathcal{H}}^{-1}).$$

Let $M_{\mathcal{H}} \in \mathbb{R}^{N \times q_{\mathcal{H}}}$ span $\text{col}\{(D_{\mathcal{H}}^\infty)^\top\}$ and set $\hat{s}_0^\dagger := M_{\mathcal{H}} \hat{c}_{M_{\mathcal{H}}}$. Then

$$\mathbb{E}[R_{\mathcal{H}}(\hat{s}_0^\dagger) - R_{\mathcal{H}}(s_{\mathcal{H}})] = \frac{q_{\mathcal{H}}}{2n_{\mathcal{H}}} + o(n_{\mathcal{H}}^{-1}).$$

- (iii) Under the remaining conditions of Theorem 4, with connectivity of $\mathcal{G}_{\mathcal{H}}^\infty$ replaced by $\text{rank}(D_{\mathcal{H}}^\infty M) = d$, the conclusion (14) and the consistency of the sandwich estimator (D.8) continue to hold.

Proof of Theorem C.5. For (i), equality of population probabilities gives $D_{\mathcal{H}}^\infty M(c_{\mathcal{H}} - \tilde{c}_{\mathcal{H}}) = 0$, so the rank condition gives $c_{\mathcal{H}} = \tilde{c}_{\mathcal{H}}$. If $C_{\mathcal{H}} > 1$, then $\dim \ker(D_{\mathcal{H}}^\infty) = C_{\mathcal{H}}$, which contains a nonzero centered direction.

For (ii), $\text{rank}(D_{\mathcal{H}}^\infty M) = d$ is exactly (C.4) for M , so exact calibration and Lemma C.3 give the first expansion. For $M_{\mathcal{H}}$, $D_{\mathcal{H}}^\infty$ is injective on $\text{col}(M_{\mathcal{H}})$ and the restricted model reproduces $D_{\mathcal{H}}^\infty s_{\mathcal{H}}$; applying Lemma C.3 with dimension $q_{\mathcal{H}}$ gives the second expansion.

For (iii), Lemma D.3 gives $\mathcal{H}_\lambda \succ 0$ under $\text{rank}(D_{\mathcal{H}}^\infty M) = d$, and Lemma D.4 then allows the proof of Theorem 4 to proceed unchanged. $\square$

If comparisons cover only $\mathcal{C} \subset [N]$ and its induced graph is connected, then $\text{rank}(D_{\mathcal{H}}^\infty M) = d$ is equivalent to $\text{rank}(P_{\mathcal{C}} M_{\mathcal{C}}) = d$, where $P_{\mathcal{C}} := I_{|\mathcal{C}|} - |\mathcal{C}|^{-1} \mathbf{1}\mathbf{1}^\top$. Hence $s_{\mathcal{H}}$ can remain identifiable on all N items even when items outside $\mathcal{C}$ receive no human comparisons.

C.4 Additional Results for Human Calibration

Effect of estimating the calibration space. The oracle analysis of Theorem 3 treats M as fixed, whereas the staged endpoint (12) estimates it from the LLM comparisons. Under exact calibration, write $H_M := M^\top L M$.

Corollary C.6 (Oracle equivalence of the staged endpoint). *Suppose the exact-calibration case of Theorem 3 holds, Proposition B.5 applies, and $n_{\mathcal{H}}, n_{\mathcal{L}} \rightarrow \infty$ with $n_{\mathcal{H}}/n_{\mathcal{L}} \rightarrow 0$. Then*

$$\|\hat{s}_\infty - \tilde{s}_\infty\|_2 = \mathcal{O}_{\mathbb{P}}(n_{\mathcal{L}}^{-1/2}), \quad \sqrt{n_{\mathcal{H}}}(\hat{s}_\infty - s_{\mathcal{H}}) \xrightarrow{d} \mathcal{N}(0, M H_M^{-1} M^\top),$$

and

$$R_{\mathcal{H}}(\hat{s}_\infty) - R_{\mathcal{H}}(\tilde{s}_\infty) = o_P(n_{\mathcal{H}}^{-1}).$$

Proof of Corollary C.6. By Proposition B.5 and smoothness of $S \mapsto P_{\text{col}(M_\theta)}$,

$$\|P_{\text{col}(\widehat{M}_\infty)} - P_{\text{col}(M)}\|_F = \mathcal{O}_{\mathbb{P}}(n_L^{-1/2}).$$

Local Lipschitz continuity of the restricted human minimizer therefore gives $\|\widehat{s}_\infty - \widetilde{s}_\infty\|_2 = \mathcal{O}_{\mathbb{P}}(n_L^{-1/2})$. Since $n_H/n_L \rightarrow 0$, $\sqrt{n_H}(\widehat{s}_\infty - \widetilde{s}_\infty) \xrightarrow{P} 0$, while Lemma C.3 gives the stated limit law for $\widetilde{s}_\infty$.

Finally, Lemma C.3 gives $\|\widetilde{s}_\infty - s_H\|_2 = \mathcal{O}_{\mathbb{P}}(n_H^{-1/2})$. Since $\nabla R_H(s_H) = 0$ and $\nabla^2 R_H$ is bounded,

$$|R_H(\widehat{s}_\infty) - R_H(\widetilde{s}_\infty)| = \mathcal{O}_{\mathbb{P}}\left((n_H n_L)^{-1/2} + n_L^{-1}\right) = o_P(n_H^{-1}).$$

The same limit is reached by $\widehat{s}_\lambda$ along any $\lambda \rightarrow \infty$ by Corollary D.5. □

Ranking implication. Let $c_M^* \in \operatorname{argmin}_c R_H(Mc)$ and $e := Mc_M^* - s_H$. If $\min_{i \neq j} |s_{Hi} - s_{Hj}| > 2\|e\|_\infty$, then Mc_M^* and s_H induce the same ranking; under (4), $e = 0_N$.

D Estimation, Inference, and Adaptive Tuning

D.1 Reduced score-space representation and regularity

The criterion in (10), restricted to the space $\mathcal{M}_r$ in (13), is

$$Q_\lambda(s, S, b) := \ell_H(s) + \lambda \ell_L(S, b), \quad (s, S, b) \in \mathcal{M}_r. \quad (\text{D.1})$$

Under Condition 3,

$$S = [\gamma, U][\mu, V]^\top, \quad \text{rank}(S) = r + 1, \quad \text{row}(S) = \text{col}(W),$$

with

$$S1_N = 0_K, \quad S^\top 1_K = (1_K^\top \gamma) \mu \neq 0_N.$$

Conversely, let $S1_N = 0_K$, $\text{rank}(S) = r + 1$, and $S^\top 1_K \neq 0_N$, and define

$$\mu := \sqrt{N} \frac{S^\top 1_K}{\|S^\top 1_K\|_2}, \quad \gamma := N^{-1} S \mu, \quad R := S - \gamma \mu^\top.$$

Then $1_N^\top \mu = 0$, $1_K^\top \gamma > 0$, $1_K^\top R = 0_N^\top$, $R\mu = 0_K$, and $\text{rank}(R) = r$. If $R = P\Sigma Q^\top$ is a compact singular-value decomposition, setting $V := \sqrt{N} Q$ and $U := N^{-1/2} P\Sigma$ yields a factorization satisfying Condition 3. Thus every admissible S has a regular factorization, unique only up to orthogonal rotation of (U, V) .

Lemma D.1 (Factorized and reduced criterion equivalence). *For every $\lambda > 0$, the map*

$$\mathcal{T}_M(c, \theta) := (M_\theta c, S_\theta, b), \quad (c, \theta) \in \mathbb{R}^d \times \Theta_r,$$

is onto $\mathcal{M}_r$ and satisfies

$$Q_\lambda(M_\theta c, S_\theta, b) = Q_\lambda\{\mathcal{T}_M(c, \theta)\}.$$

Consequently,

$$\inf_{\substack{c \in \mathbb{R}^d \\ \theta \in \Theta_r}} Q_\lambda(M_\theta c, S_\theta, b) = \inf_{(s, S, b) \in \mathcal{M}_r} Q_\lambda(s, S, b).$$

If $(\hat{c}_\lambda, \hat{\theta}_\lambda)$ minimizes (11), then

$$(M_{\hat{\theta}_\lambda} \hat{c}_\lambda, S_{\hat{\theta}_\lambda}, \hat{b}_\lambda)$$

is a minimizer of (D.1). Conversely, every minimizer of (D.1) admits a regular factorization with the same calibrated score and objective value.

Proof of Lemma D.1. Fix $\theta \in \Theta_r$. The constraints defining Θ_r give $S_\theta 1_N = 0_K$, $S_\theta = [\gamma, U]W_\theta^\top$, and, since $1_K^\top U = 0^\top$, $S_\theta^\top 1_K = (1_K^\top \gamma) \mu \neq 0_N$ because $1_K^\top \gamma > 0$ and $N^{-1} \|\mu\|_2^2 = 1$. Moreover V has rank r and $\mu^\top V = 0$, so $\text{rank}(W_\theta) = r + 1$ and $\text{row}(S_\theta) = \text{col}(W_\theta)$, both spaces having dimension $r + 1$. Hence $\mathcal{C}_M(S_\theta) = \text{col}(M_\theta)$ and $\text{rank}(M_\theta) = d$ in either case, so $\mathcal{T}_M$ maps into $\mathcal{M}_r$. Conversely, for $(s, S, b) \in \mathcal{M}_r$, the reconstruction above gives $\theta \in \Theta_r$ with $S_\theta = S$ and $\text{col}(M_\theta) = \mathcal{C}_M(S)$, so $s = M_\theta c$ for some $c \in \mathbb{R}^d$ and $\mathcal{T}_M$ is onto. By (10),

$$Q_\lambda(M_\theta c, S_\theta, b) = \ell_H(M_\theta c) + \lambda \ell_L(S_\theta, b) = Q_\lambda\{\mathcal{T}_M(c, \theta)\},$$

so the two infima agree by surjectivity. Comparison with every factorization shows that any global minimizer of (11) maps to a minimizer of (D.1), and the converse follows from surjectivity and objective preservation. $\square$

The set $\mathcal{M}_r$ is locally a smooth manifold: after imposing $S1_N = 0_K$, the exact-rank constraint is locally the rank- $(r + 1)$ manifold in $\mathbb{R}^{K \times (N-1)}$, while $S^\top 1_K \neq 0_N$ is open.

Explicit coordinates. For analysis, use the over-parameterization

$$\begin{aligned}\varphi = (A, B, c, b) &\mapsto (s, S, b) = (s_M(A, B, c), AB^\top, b), \\ s_M(A, B, c) &:= \begin{cases} c\sqrt{N} \frac{BA^\top 1_K}{\|BA^\top 1_K\|_2}, & M = \mu, \\ Bc, & M = W, \end{cases} \\ A \in \mathbb{R}^{K \times (r+1)}, \quad B \in \mathbb{R}^{N \times (r+1)}, \quad B^\top 1_N &= 0_{r+1}, \end{aligned} \quad (\text{D.2})$$

with A and B full column rank, $A^\top 1_K \neq 0_{r+1}$, $c \in \mathbb{R}^d$, and $b \in \mathbb{R}^K$. Its image is $\mathcal{M}_r$, and its fibres are exactly the gauge orbits

$$(A, B, c) \mapsto \begin{cases} (AG^{-\top}, BG, c), & M = \mu, \\ (AG^{-\top}, BG, G^{-1}c), & M = W, \end{cases} \quad G \in GL(r+1). \quad (\text{D.3})$$

Therefore

$$\dim \mathcal{M}_r = (r+1)(K+N-r-2) + d + K. \quad (\text{D.4})$$

Thus the regular fibre of (D.2) has dimension $(r+1)^2$, the dimension of $GL(r+1)$. For computation, we use the restriction

$$A^\top 1_K = Ke_1,$$

which remains a surjective local parameterization of $\mathcal{M}_r$ and restricts the gauge in (D.3) to $\{G \in GL(r+1) : Ge_1 = e_1\}$, of dimension $r(r+1)$. For $M = \mu$, its scalar calibration coordinate is equivalently written as α with $s = \alpha Be_1$.

Lemma D.2 (Gauge-invariant sandwich and trace). *Fix a local minimizer of q_λ on $\mathcal{M}_r$, and let ζ be any smooth chart around it in which the Hessian $H_\zeta := \nabla_\zeta^2 q_\lambda$ is nonsingular. Let J_ζ be the covariance of the individual human scores in that chart. Let φ be coordinates in (D.2), or in any smooth restriction that remains a submersion onto $\mathcal{M}_r$, including the computational restriction $A^\top 1_K = Ke_1$, and set $H_\varphi := \nabla_\varphi^2 q_\lambda(\varphi)$, J_φ the same score covariance, and $\dot{s}_\varphi := \partial s / \partial \varphi^\top$. Then, with Moore-Penrose inverses,*

$$\text{tr}(H_\varphi^+ J_\varphi) = \text{tr}(H_\zeta^{-1} J_\zeta), \quad \dot{s}_\varphi H_\varphi^+ J_\varphi H_\varphi^+ \dot{s}_\varphi^\top = \dot{s}_\zeta H_\zeta^{-1} J_\zeta H_\zeta^{-1} \dot{s}_\zeta^\top.$$

Consequently the GACV penalty in (15) and the sandwich (D.8) can be evaluated in either over-parameterized chart, and neither depends on the chart.

Proof of Lemma D.2. Let $T := \partial \zeta / \partial \varphi^\top$ at φ . Because the chosen parameterization is a submersion onto $\mathcal{M}_r$, T has full row rank. Every quantity involved is a function of (s, S, b) alone, so $\nabla_\varphi = T^\top \nabla_\zeta$; since $\nabla_\zeta q_\lambda = 0$ at the point, the term involving the second derivative of ζ drops and

$$H_\varphi = T^\top H_\zeta T, \quad J_\varphi = T^\top J_\zeta T, \quad \dot{s}_\varphi = \dot{s}_\zeta T.$$

Take a thin QR factorization $T^\top = QR$ with $Q^\top Q = I$ and R invertible, and set $G := RH_\zeta R^\top \succ 0$. Then $H_\varphi = QGQ^\top$, so $H_\varphi^+ = QG^{-1}Q^\top$, while $J_\varphi = Q(RJ_\zeta R^\top)Q^\top$. Hence

$$\text{tr}(H_\varphi^+ J_\varphi) = \text{tr}(G^{-1}RJ_\zeta R^\top) = \text{tr}(R^{-\top}H_\zeta^{-1}J_\zeta R^\top) = \text{tr}(H_\zeta^{-1}J_\zeta).$$

Since $TQ = R^\top$, the same substitution gives $\dot{s}_\varphi H_\varphi^+ J_\varphi H_\varphi^+ \dot{s}_\varphi^\top = \dot{s}_\zeta H_\zeta^{-1} J_\zeta H_\zeta^{-1} \dot{s}_\zeta^\top$. $\square$

Regularity and endpoints. Neither Θ_r nor $\mathcal{M}_r$ is closed because rank may drop and $S^\top 1_K \neq 0_N$ is an open restriction. The item-side normalization in Condition 3 removes scale escape; under a finite identifying design with limiting probabilities in $(0, 1)$, the remaining asymptotic failure modes are rank drop and separation. At $\lambda = 0$, the score-level problem reduces to the unrestricted centered human BTL estimator $\hat{s}_0$, while its factor coordinates are unidentified. For finite λ , the human likelihood can perturb (S, b) , so we report judge diagnostics from the LLM-only fit $\hat{\theta}_\infty$ and reserve $\hat{s}_\lambda$ for the human target.

For tuning and inference, Lemma D.1 lets us work on $\mathcal{M}_r$, while Lemmas D.3 and D.4 give the required local regularity.

Weighted information and consistency.

Lemma D.3 (When the weighted information is nonsingular). *Suppose the limiting LLM comparison design satisfies Condition 2. Then $\ker(\mathcal{I}_L)$ is exactly the d -dimensional fibre along which s moves inside $\mathcal{C}_M(S) = \text{col}(M)$ with (S, b) fixed, and $\mathcal{I}_H$ restricted to that fibre is congruent to $M^\top LM$. Consequently, for every $\lambda > 0$,*

$$\mathcal{H}_\lambda \succ 0 \iff \text{rank}(D_H^\infty M) = d.$$

Proof of Lemma D.3. For a tangent direction v , write $(\dot{s}_v, \dot{S}_v, \dot{b}_v)$ for its score-level differential. Since $\mathcal{I}_L$ is the Hessian of the limiting LLM risk, $v^\top \mathcal{I}_L v = 0$ if and only if the LLM linear predictor is unchanged, so

$$D_k \dot{S}_{v,k}^\top + z_k \dot{b}_{v,k} = 0$$

on the limiting support for every k . By Condition 2 and row centering, this implies $\dot{S}_v = 0$ and $\dot{b}_v = 0$. With (S, b) fixed, the remaining directions are $\dot{s}_v = M \dot{c}_v$, giving a d -dimensional fibre. If Π is a basis of this fibre, then $\dot{s}_0 \Pi = MC$ for some invertible C , and

$$\Pi^\top \mathcal{I}_H \Pi = C^\top M^\top LMC.$$

Hence $\mathcal{H}_\lambda \succ 0$ if and only if $M^\top LM \succ 0$. By (C.6),

$$u^\top M^\top LM u = \sum_{(i,j) \in \Omega_H^\infty} \rho_{ij} p_{Hij} (1 - p_{Hij}) \{(e_i - e_j)^\top M u\}^2,$$

which is positive for every $u \neq 0$ exactly when $\text{rank}(D_H^\infty M) = d$. $\square$

The condition Condition 2 is sufficient but not necessary here: if it fails, a direction invisible to the LLM likelihood may still move $\mathcal{C}_M(S)$ and hence be detected by the human likelihood.

Effective dimension. For $A, B \succeq 0$ with $A + \lambda B \succ 0$, define

$$\text{df}(\lambda) := \text{tr}\{(A + \lambda B)^{-1} A\}.$$

Writing $G_\lambda = (A + \lambda B)^{-1}$ gives

$$\text{df}'(\lambda) = -\text{tr}(G_\lambda B G_\lambda A) \leq 0,$$

with $\text{df}(0^+) = \text{rank}(A)$ and $\text{df}(\infty) = \dim \ker(B)$. Taking $A = \mathcal{I}_H$ and $B = \mathcal{I}_L$, Lemma D.3 gives $\dim \ker(\mathcal{I}_L) = d$, so the effective dimension decreases from $\text{rank}(\mathcal{I}_H)$ to d , and from $N - 1$ to d under the connected-design setting of Theorem 4 when $\text{rank}(\dot{s}_0) = N - 1$.

Lemma D.4 (Regularity and consistency of the weighted estimator). *Fix $\lambda \in (0, \infty)$ and N, K, r . Under model (1), (3), and (4), Conditions 1 and 3, let the human comparisons follow the independent-draw scheme of Section 4.2, let $n_H, n_L \rightarrow \infty$, and suppose $n_k/n_L \rightarrow \kappa_k > 0$ and $n_{kij}^{(a)}/n_k \rightarrow \rho_{kij}^{(a)}$, with each limiting within-judge support satisfying Condition 2. Assume $\text{rank}(D_H^\infty M) = d$ and that all active limiting comparison probabilities lie in $(0, 1)$. Then the population criterion*

$$R_H(s) + \lambda R_L(S, b)$$

has the unique minimizer (s_H, S, b) over $\mathcal{M}_r$, and its value is strictly separated from the minimum along any sequence approaching the boundary of $\mathcal{M}_r$. With probability tending to one, the empirical reduced criterion has a finite global minimizer in $\mathcal{M}_r$, every measurable sequence of such minimizers is consistent, and its Hessian in any smooth local coordinate on $\mathcal{M}_r$ is positive definite. Moreover, the score-level image of every global minimizer $(\hat{c}_\lambda, \hat{\theta}_\lambda)$ of (11) is a global minimizer of the reduced criterion.

Proof of Lemma D.4. By correct specification,

$$R_H(s) - R_H(s_H) \geq 0, \quad R_L(S, b) - R_L(S_0, b_0) \geq 0.$$

Under Condition 2, equality in the second display requires $(S, b) = (S_0, b_0)$. At (S_0, b_0) , exact calibration gives $s, s_H \in \mathcal{C}_M(S_0) = \text{col}(M)$, while equality in the human risk implies $D_H^\infty(s - s_H) = 0$. Hence $\text{rank}(D_H^\infty M) = d$ gives $s = s_H$, proving uniqueness of the population minimizer.

Because the comparison support is finite and all active probabilities are interior, with probability tending to one every active cell contains both outcomes. The empirical logistic losses are then coercive in their active linear predictors. Together with Condition 2, this bounds (S, b) on every relevant empirical sublevel. Moreover, the unrestricted minimum human loss converges to $R_H(s_H)$. Since the population LLM risk is separated from its minimum outside every neighborhood of (S_0, b_0) , uniform convergence on bounded predictor sets implies that every empirical global minimizer has (S, b) in an arbitrarily small neighborhood of (S_0, b_0) with probability tending to one.

By continuity of $\mathcal{C}_M(S)$ and $\text{rank}(D_H^\infty M) = d$, on such a neighborhood D_H^∞ is uniformly injective on $\mathcal{C}_M(S)$. The human loss is therefore coercive in the calibration coordinates there, so the relevant empirical sublevel is compact. Uniform convergence and uniqueness of the population minimizer give consistency and existence of a finite minimizer in $\mathcal{M}_r$.

Finally, Lemma D.3 gives $\mathcal{H}_\lambda \succ 0$ at the truth. Consistency and local uniform convergence of the empirical Hessian then imply positive definiteness of the Hessian in any smooth local coordinate with probability tending to one. The final claim follows from Lemma D.1. $\square$

D.2 Proof of Theorem 4

We work in the local coordinate ζ introduced in Section 5.2. Let ζ_0 denote the coordinate of the true reduced parameter (s_H, S, b) . Let $\hat{\zeta}_\lambda$ denote the local coordinate of the score-level image $(\hat{s}_\lambda, \hat{S}_\lambda, \hat{b}_\lambda)$ of the factorized estimator in (11). By Lemmas D.1 and D.4, with probability tending to one this image is a regular global minimizer of the reduced criterion (D.1) and $\hat{\zeta}_\lambda \xrightarrow{P} \zeta_0$. Fix a relatively compact smooth chart neighborhood of ζ_0 ; the finite-support logistic criterion has uniformly bounded derivatives there. Throughout this subsection, consider any sequence satisfying $n_H, n_L \rightarrow \infty$ and $n_H/n_L \rightarrow \tau \in [0, \infty)$.

For the LLM sample, expand each binomial count into its individual Bernoulli comparisons. For every comparison cell in the limiting support,

$$\frac{n_{kij}^{(a)}}{n_L} = \frac{n_k}{n_L} \cdot \frac{n_{kij}^{(a)}}{n_k} \longrightarrow \kappa_k \rho_{kij}^{(a)},$$

where $\sum_k \kappa_k = 1$ and $\sum_{(i,j,a)} \rho_{kij}^{(a)} = 1$ for each k , so the cell weights $\kappa_k \rho_{kij}^{(a)}$ sum to one, and each limiting within-judge support satisfies Condition 2. Writing

$$\eta_{kij}^{(a)}(S, b) = S_{ki} - S_{kj} + ab_k,$$

the limiting population LLM loss is

$$R_L(S, b) = \sum_{k=1}^K \sum_{i < j} \sum_{a \in \{-1, 1\}} \kappa_k \rho_{kij}^{(a)} \left[\log \left\{ 1 + \exp \left(\eta_{kij}^{(a)}(S, b) \right) \right\} - p_{kij}^{(a)} \eta_{kij}^{(a)}(S, b) \right], \quad (\text{D.5})$$

where terms with $\kappa_k \rho_{kij}^{(a)} = 0$ are omitted.

Proof of Theorem 4. For the t -th human comparison between items (i_t, j_t) , define its negative log-likelihood contribution as

$$h_t(\zeta) := \log \{ 1 + \exp(\eta_{H, i_t j_t}(\zeta)) \} - Y_t \eta_{H, i_t j_t}(\zeta). \quad (\text{D.6})$$

For the u -th LLM comparison, corresponding to judge k_u , items (i_u, j_u) , and order $a_u \in \{-1, 1\}$, define

$$l_u(\zeta) := \log \left\{ 1 + \exp \left(\eta_{k_u i_u j_u}^{(a_u)}(\zeta) \right) \right\} - Y_u \eta_{k_u i_u j_u}^{(a_u)}(\zeta). \quad (\text{D.7})$$

Both are expressed through the reduced local coordinate ζ . Then

$$q_\lambda(\zeta) = \frac{1}{n_H} \sum_{t=1}^{n_H} h_t(\zeta) + \frac{\lambda}{n_L} \sum_{u=1}^{n_L} l_u(\zeta).$$

Part 1: Asymptotic normality. Under the human BTL model, the LLM order-effect model, and the calibration restriction (4), ζ_0 correctly specifies both likelihood components, so the two population scores vanish separately at ζ_0 . The information identity gives

$$\text{Var} \{ \nabla_{\zeta} h_t(\zeta_0) \} = \mathcal{I}_H$$

for the human observations, while the limiting LLM design gives

$$\frac{1}{n_L} \sum_{u=1}^{n_L} \text{Var} \{ \nabla_{\zeta} l_u(\zeta_0) \} \longrightarrow \mathcal{I}_L.$$

Both $\mathcal{I}_H$ and $\mathcal{I}_L$ are positive semidefinite, and $\mathcal{H}_{\lambda} \succ 0$ by Lemma D.3, since connectivity of $\mathcal{G}_H^{\infty}$ makes L positive definite on the centered space and hence gives $\text{rank}(D_H^{\infty} M) = d$.

Because the human and LLM samples are independent,

$$\sqrt{n_H} \nabla_{\zeta} q_{\lambda}(\zeta_0) = \frac{1}{\sqrt{n_H}} \sum_{t=1}^{n_H} \nabla_{\zeta} h_t(\zeta_0) + \lambda \sqrt{\frac{n_H}{n_L}} \frac{1}{\sqrt{n_L}} \sum_{u=1}^{n_L} \nabla_{\zeta} l_u(\zeta_0).$$

The human term converges to $\mathcal{N}(0, \mathcal{I}_H)$. For the LLM triangular array, the individual score vectors are uniformly bounded and their average covariance converges to $\mathcal{I}_L$ under the limiting cell proportions, so the Lindeberg-Feller CLT gives

$$\frac{1}{\sqrt{n_L}} \sum_{u=1}^{n_L} \nabla_{\zeta} l_u(\zeta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_L).$$

Therefore, if $\tau > 0$, independence and the joint central limit theorem give

$$\sqrt{n_H} \nabla_{\zeta} q_{\lambda}(\zeta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_H + \lambda^2 \tau \mathcal{I}_L).$$

If $\tau = 0$, then

$$\lambda \sqrt{\frac{n_H}{n_L}} \frac{1}{\sqrt{n_L}} \sum_{u=1}^{n_L} \nabla_{\zeta} l_u(\zeta_0) = o_{\mathbb{P}}(1),$$

so the same conclusion holds with $\tau = 0$. Hence, for any sequence with $n_H/n_L \rightarrow \tau \in [0, \infty)$,

$$\sqrt{n_H} \nabla_{\zeta} q_{\lambda}(\zeta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{J}_{\lambda, \tau}).$$

Because the comparison support is finite and $\hat{\zeta}_{\lambda} \xrightarrow{p} \zeta_0$, the local uniform law of large numbers gives

$$\nabla_{\zeta}^2 q_{\lambda}(\tilde{\zeta}_{\lambda}) \xrightarrow{p} \mathcal{I}_H + \lambda \mathcal{I}_L = \mathcal{H}_{\lambda}$$

for any intermediate point $\tilde{\zeta}_{\lambda}$ between $\hat{\zeta}_{\lambda}$ and ζ_0 . A Taylor expansion of the first-order condition therefore gives

$$0 = \nabla_{\zeta} q_{\lambda}(\zeta_0) + \{ \mathcal{H}_{\lambda} + o_{\mathbb{P}}(1) \} (\hat{\zeta}_{\lambda} - \zeta_0),$$

and hence

$$\sqrt{n_H}(\hat{\zeta}_{\lambda} - \zeta_0) = -\mathcal{H}_{\lambda}^{-1} \sqrt{n_H} \nabla_{\zeta} q_{\lambda}(\zeta_0) + o_{\mathbb{P}}(1).$$

It follows that

$$\sqrt{n_H}(\hat{\zeta}_{\lambda} - \zeta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{H}_{\lambda}^{-1} \mathcal{J}_{\lambda, \tau} \mathcal{H}_{\lambda}^{-1}).$$

Applying the delta method to $\zeta \mapsto s(\zeta)$ gives

$$\sqrt{n_H}(\hat{s}_{\lambda} - s_H) \xrightarrow{d} \mathcal{N}(0, \dot{s}_0 \mathcal{H}_{\lambda}^{-1} \mathcal{J}_{\lambda, \tau} \mathcal{H}_{\lambda}^{-1} \dot{s}_0^{\top}).$$

Under (4), $s_H = s(\zeta_0) = M_{C_H}$, which proves (14).

Part 2: Variance estimator. Under a smooth nonsingular change of local coordinates, $\mathcal{H}_\lambda$, $\mathcal{J}_{\lambda,\tau}$, and $\dot{s}_0$ transform covariantly, so the displayed covariance is unchanged. When $\tau = 0$, $\mathcal{J}_{\lambda,0} = \mathcal{I}_H$, giving the abundant-LLM expression in Theorem 4. For completeness, define the empirical sandwich covariance explicitly. At $\hat{\zeta}_\lambda$, let

$$g_{Ht} := \nabla_\zeta h_t(\hat{\zeta}_\lambda), \quad \bar{g}_H := \frac{1}{n_H} \sum_{t=1}^{n_H} g_{Ht},$$

and

$$g_{Lu} := \nabla_\zeta l_u(\hat{\zeta}_\lambda), \quad \bar{g}_L := \frac{1}{n_L} \sum_{u=1}^{n_L} g_{Lu}.$$

Define

$$\hat{\mathcal{V}}_H := \frac{1}{n_H} \sum_{t=1}^{n_H} (g_{Ht} - \bar{g}_H)(g_{Ht} - \bar{g}_H)^\top,$$

$$\hat{\mathcal{V}}_L := \frac{1}{n_L} \sum_{u=1}^{n_L} (g_{Lu} - \bar{g}_L)(g_{Lu} - \bar{g}_L)^\top,$$

and

$$\hat{\mathcal{H}}_\lambda := \nabla_\zeta^2 q_\lambda(\hat{\zeta}_\lambda), \quad \hat{\mathcal{J}}_\lambda := \hat{\mathcal{V}}_H + \lambda^2 \frac{n_H}{n_L} \hat{\mathcal{V}}_L.$$

Writing

$$\hat{s}_\lambda := \nabla_\zeta s(\hat{\zeta}_\lambda),$$

define

$$\hat{\Sigma}_{\lambda,s} := \hat{s}_\lambda \hat{\mathcal{H}}_\lambda^{-1} \hat{\mathcal{J}}_\lambda \hat{\mathcal{H}}_\lambda^{-1} \hat{s}_\lambda^\top. \quad (\text{D.8})$$

By Lemma D.4, finite comparison support, and the limiting-design condition,

$$\hat{\mathcal{H}}_\lambda \xrightarrow{p} \mathcal{H}_\lambda, \quad \hat{\mathcal{V}}_H \xrightarrow{p} \mathcal{I}_H, \quad \hat{\mathcal{V}}_L \xrightarrow{p} \mathcal{I}_L, \quad \hat{s}_\lambda \xrightarrow{p} \dot{s}_0.$$

Since $n_H/n_L \rightarrow \tau$ and $\mathcal{H}_\lambda$ is positive definite,

$$\hat{\Sigma}_{\lambda,s} \xrightarrow{p} \Sigma_{\lambda,s,\tau}$$

by the continuous mapping theorem. Thus $n_H^{-1} \hat{\Sigma}_{\lambda,s}$ consistently estimates the first-order covariance of $\hat{s}_\lambda$. The same change-of-coordinate calculation as above shows that (D.8) is invariant to the chosen smooth local coordinate.

Part 3: Uniformity and infinity. The uniform linearization, sandwich consistency, and $\lambda \rightarrow \infty$ limit follow from Corollary D.5. $\square$

Corollary D.5 (Uniformity in the calibration weight). *Under the conditions of Theorem 4 with $n_H/n_L \rightarrow 0$, for every $\lambda_0 > 0$,*

$$\sup_{\lambda \geq \lambda_0} \left\| \sqrt{n_H}(\hat{s}_\lambda - s_H) + \dot{s}_0 \mathcal{H}_\lambda^{-1} \sqrt{n_H} \nabla_\zeta \ell_H \{s(\zeta_0)\} \right\|_2 = o_{\mathbb{P}}(1), \quad \sup_{\lambda \geq \lambda_0} \left\| \hat{\Sigma}_{\lambda,s} - \Sigma_{\lambda,s,0} \right\| = o_{\mathbb{P}}(1),$$

and $\Sigma_{\lambda,s,0} \rightarrow M H_M^{-1} M^\top$ as $\lambda \rightarrow \infty$. Hence, if $\lambda_n \rightarrow \lambda_* \in [\lambda_0, \infty)$, (14) holds with covariance $\Sigma_{\lambda_*,s,0}$; if $\lambda_n \rightarrow \infty$, its limit is that of Corollary C.6.

Proof of Corollary D.5. By coordinate invariance, use the product chart $\zeta = (\xi, c)$, where ξ is a smooth chart of the regular structured (S, b) model as in Proposition B.5 and $s = M(\xi)c$ for a smooth local basis $M(\xi)$ of $\mathcal{C}_M\{S(\xi)\}$ with $M(\xi_0) = M$. In this chart the LLM loss depends on ξ alone, so its score and Hessian have zero c -blocks at every ζ , $\mathcal{I}_L = \text{diag}(\mathcal{I}_\xi, 0)$ with $\mathcal{I}_\xi \succ 0$, and $\partial s / \partial c = M$ at ζ_0 . Write $\mathcal{I}_H = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix}$ with $C = M^\top L M \succ 0$ (Lemma D.3). Block inversion of $\mathcal{H}_\lambda$ gives, for a constant c_0 depending on λ_0 only,

$$\mathcal{H}_\lambda^{-1} = \begin{pmatrix} 0 & 0 \\ 0 & C^{-1} \end{pmatrix} + O(\lambda^{-1}), \quad \sup_{\lambda \geq \lambda_0} \left\| \lambda \mathcal{H}_\lambda^{-1} \text{diag}(E, 0) \right\| \leq c_0 \|E\| \quad \text{for every } E. \quad (\text{D.9})$$

In the proof of Lemma D.4, the inequality $\lambda\{\ell_L(\hat{\xi}_\lambda) - \ell_L(\xi_0)\} \leq \ell_H(s_H) - \inf \ell_H = o_{\mathbb{P}}(1)$ holds with λ replaced by λ_0 for all $\lambda \geq \lambda_0$, and the remaining steps do not involve λ , so $\sup_{\lambda \geq \lambda_0} \|\hat{\xi}_\lambda - \xi_0\| = o_{\mathbb{P}}(1)$. The first-order condition reads $0 = \nabla q_\lambda(\zeta_0) + \{\mathcal{H}_\lambda + E_H + \lambda \text{diag}(E_\xi, 0)\}(\hat{\xi}_\lambda - \zeta_0)$, where $E_H = o_{\mathbb{P}}(1)$ uniformly, and $E_\xi = o(1) + O(\|\hat{\xi}_\lambda - \zeta_0\|)$ because the logistic Hessian of ℓ_L is free of the outcomes. Multiplying by $\mathcal{H}_\lambda^{-1}$ and using (D.9), $\sqrt{n_H}(\hat{\xi}_\lambda - \zeta_0) = -\{I + o_{\mathbb{P}}(1)\}\mathcal{H}_\lambda^{-1}\sqrt{n_H}\nabla q_\lambda(\zeta_0)$ uniformly in $\lambda \geq \lambda_0$. Here $\sqrt{n_H}\nabla q_\lambda(\zeta_0) = \sqrt{n_H}\nabla \ell_H\{s(\zeta_0)\} + \lambda\sqrt{n_H/n_L}(Z_L^\top, 0)^\top$ with $Z_L = \mathcal{O}_{\mathbb{P}}(1)$, and (D.9) gives $\mathcal{H}_\lambda^{-1}(\lambda Z_L^\top, 0)^\top = \mathcal{O}_{\mathbb{P}}(1)$ uniformly, so the LLM term is $\mathcal{O}_{\mathbb{P}}(\sqrt{n_H/n_L}) = o_{\mathbb{P}}(1)$; the delta method with $\hat{s}_\lambda \xrightarrow{P} \dot{s}_0$ uniformly proves the first display. For the second, $\hat{\mathcal{H}}_\lambda^{-1} - \mathcal{H}_\lambda^{-1} = o_{\mathbb{P}}(1)$ uniformly by the same expansion of $\hat{\mathcal{H}}_\lambda = \mathcal{H}_\lambda + E_H + \lambda \text{diag}(E_\xi, 0)$, and the LLM part of the meat contributes $\hat{\mathcal{H}}_\lambda^{-1}\lambda^2(n_H/n_L)\text{diag}(\hat{V}_\xi, 0)\hat{\mathcal{H}}_\lambda^{-1} = \mathcal{O}_{\mathbb{P}}(n_H/n_L)$ uniformly by (D.9). Finally, the first part of (D.9) gives $\Sigma_{\lambda, s, 0} \rightarrow \dot{s}_0 \text{diag}(0, C^{-1})\dot{s}_0^\top = MC^{-1}M^\top$. $\square$

Local departures from exact calibration. Theorem 4 assumes (4) exactly. The following proposition gives the first-order limit under an $n_H^{-1/2}$ departure.

Proposition D.6 (Inference under local misalignment). *Assume the conditions of Theorem 4 and $c_H \neq 0$, with (4) replaced by*

$$s_H^{(n_H)} = Mc_H + n_H^{-1/2}h, \quad h \in \mathbb{R}^N, \quad 1_N^\top h = 0. \quad (\text{D.10})$$

Then

$$\sqrt{n_H}(\hat{s}_\lambda - s_H^{(n_H)}) \xrightarrow{d} \mathcal{N}(-(I_N - A_\lambda)h, \Sigma_{\lambda, s, \tau}), \quad A_\lambda := \dot{s}_0 \mathcal{H}_\lambda^{-1} \dot{s}_0^\top L, \quad (\text{D.11})$$

with L the comparison Laplacian (C.6). On the centered subspace, A_λ is self-adjoint in the L -inner product, its ordered eigenvalues lie in $[0, 1]$ and are nonincreasing in λ , and $A_\lambda h = h$ for $h \in \text{col}(M)$. As $\lambda \rightarrow 0^+$ it tends to the identity on centered vectors, while as $\lambda \rightarrow \infty$ it tends to $P_M^L := M(M^\top LM)^{-1}M^\top L$.

Proof of Proposition D.6. Keep ζ_0 at the coordinate of (Mc_H, S, b) , which under (D.10) is no longer the true human score. Only the human component of the criterion changes. Writing $p_{Hij}^{(n_H)} = \sigma\{(Mc_H)_i - (Mc_H)_j + n_H^{-1/2}(h_i - h_j)\}$ and expanding the logistic function about $n_H^{-1/2} = 0$,

$$\begin{aligned} \nabla_s R_H^{(n_H)}(Mc_H) &= - \sum_{(i,j) \in \Omega_H^\infty} \rho_{ij} \left\{ p_{Hij}^{(n_H)} - \sigma((Mc_H)_i - (Mc_H)_j) \right\} (e_i - e_j) \\ &= -n_H^{-1/2}Lh + O(n_H^{-1}), \end{aligned}$$

by the definition (C.6) of L . The LLM component is unaffected, so the chain rule gives $\sqrt{n_H}\mathbb{E}\{\nabla_\zeta q_\lambda(\zeta_0)\} \rightarrow -\dot{s}_0^\top Lh$, while the score covariance is unchanged at first order because $p_{Hij}^{(n_H)} \rightarrow p_{Hij}$. Hence $\sqrt{n_H}\nabla_\zeta q_\lambda(\zeta_0) \xrightarrow{d} \mathcal{N}(-\dot{s}_0^\top Lh, \mathcal{J}_{\lambda, \tau})$, and repeating the Taylor expansion above,

$$\sqrt{n_H}(\hat{\xi}_\lambda - \zeta_0) \xrightarrow{d} \mathcal{N}(\mathcal{H}_\lambda^{-1}\dot{s}_0^\top Lh, \mathcal{H}_\lambda^{-1}\mathcal{J}_{\lambda, \tau}\mathcal{H}_\lambda^{-1}).$$

The delta method gives $\sqrt{n_H}(\hat{s}_\lambda - Mc_H) \xrightarrow{d} \mathcal{N}(A_\lambda h, \Sigma_{\lambda, s, \tau})$, and subtracting the deterministic $\sqrt{n_H}(s_H^{(n_H)} - Mc_H) = h$ proves (D.11). Since $c_H \neq 0$, the tangent map $\dot{s}_0$ spans the centered score space: for $M = \mu$, variation of c spans μ and variation of μ spans its centered orthogonal complement; for $M = W$, variation of c spans $\text{col}(W)$ and subspace variation of W_{c_H} spans its centered complement. Hence $\text{rank}(\dot{s}_0) = N - 1$.

The two endpoints. Set

$$B_\lambda := L^{1/2}\dot{s}_0 \mathcal{H}_\lambda^{-1} \dot{s}_0^\top L^{1/2}.$$

Since $\mathcal{I}_H = \dot{s}_0^\top L\dot{s}_0$ and $\text{rank}(\dot{s}_0) = N - 1$, $B_\lambda \rightarrow P$ as $\lambda \rightarrow 0^+$. By Lemma D.3, $\ker(\mathcal{I}_L)$ is the d -dimensional fibre mapped by $\dot{s}_0$ onto $\text{col}(M)$, so $B_\lambda \rightarrow P_{L^{1/2}M}$ and $A_\lambda \rightarrow P_M^L$ as $\lambda \rightarrow \infty$. For $h \in \text{col}(M)$, choose $v \in \ker(\mathcal{I}_L)$ with $\dot{s}_0 v = h$. Then

$$\mathcal{H}_\lambda v = \mathcal{I}_H v = \dot{s}_0^\top Lh,$$

so $A_\lambda h = h$.

Moreover,

$$\frac{\partial B_\lambda}{\partial \lambda} = -L^{1/2} \dot{s}_0 \mathcal{H}_\lambda^{-1} \mathcal{I}_L \mathcal{H}_\lambda^{-1} \dot{s}_0^\top L^{1/2} \preceq 0.$$

Since $B_\lambda \succeq 0$ and $B_\lambda \rightarrow P$ as $\lambda \rightarrow 0^+$,

$$0 \preceq B_\lambda \preceq P.$$

Thus, on the centered subspace, the ordered eigenvalues of B_λ lie in $[0, 1]$ and are nonincreasing in λ . Because

$$B_\lambda = L^{1/2} A_\lambda L^{1/2}$$

on that subspace, A_λ has the same eigenvalues. Finally,

$$A_\lambda^\top L = L A_\lambda,$$

so A_λ is self-adjoint in the L -inner product. Together with $B_\lambda \rightarrow P_{L^{1/2}M}$, exactly d centered directions remain unshrunk as $\lambda \rightarrow \infty$. $\square$

Interpretation. By Proposition D.6, the local bias vanishes as $\lambda \rightarrow 0^+$ and approaches $-(I_N - P_M^L)h$ as $\lambda \rightarrow \infty$, while $\Sigma_{\lambda,s,\tau}$ gives the corresponding first-order variance.

Corollary D.7 (Local risk at a finite calibration weight). *Under Proposition D.6 with $\tau = 0$, let $R_H^{(n_H)}$ denote the human risk under (D.10). For fixed $\lambda \in (0, \infty)$,*

$$n_H \left\{ R_H^{(n_H)}(\hat{s}_\lambda) - R_H^{(n_H)}(s_H^{(n_H)}) \right\} \xrightarrow{d} \frac{1}{2} \|(I_N - A_\lambda)h - G_\lambda\|_L^2, \quad G_\lambda \sim \mathcal{N}(0, \Sigma_{\lambda,s,0}).$$

The limiting variable has mean

$$\frac{1}{2} \|(I_N - A_\lambda)h\|_L^2 + \frac{1}{2} \text{tr}(B_\lambda^2), \quad B_\lambda := L^{1/2} \dot{s}_0 \mathcal{H}_\lambda^{-1} \dot{s}_0^\top L^{1/2}.$$

Moreover, $\text{tr}(B_\lambda^2) \leq \text{tr}(B_\lambda) = \text{df}(\lambda)$ and $d \leq \text{df}(\lambda) \leq N - 1$. As $\lambda \rightarrow 0^+$ and $\lambda \rightarrow \infty$, the limiting means approach $(N - 1)/2$ and $\delta_M + d/2$, respectively, along (D.10).

Proof of Corollary D.7. By Proposition D.6 and a second-order expansion of $R_H^{(n_H)}$ at $s_H^{(n_H)}$, whose Hessian converges to L , the scaled excess risk converges to the stated quadratic form. For $\tau = 0$, $\text{tr}(L \Sigma_{\lambda,s,0}) = \text{tr}(B_\lambda^2)$, while the effective-dimension calculation gives $\text{tr}(B_\lambda) = \text{df}(\lambda)$ and $0 \preceq B_\lambda \preceq P$. The endpoint claims follow from Proposition D.6 and Lemma C.1. $\square$

D.3 Proof of Theorem 5

Assumption A1 (GACV regularity). Condition on the complete LLM comparison sample and let $n_H \rightarrow \infty$. For each $\lambda \in \bar{\Lambda}$, the corresponding conditional population criterion has a unique interior minimizer ζ_λ^* with nonsingular Hessian and is well separated: for some neighborhood $\mathcal{U}_\lambda$ of ζ_λ^* and $\epsilon_\lambda > 0$,

$$\inf_{\zeta \notin \mathcal{U}_\lambda} q_\lambda^*(\zeta) \geq q_\lambda^*(\zeta_\lambda^*) + \epsilon_\lambda,$$

where q_λ^* is the conditional expectation of the human part of the criterion plus its fixed LLM part. The full-sample fit converges in probability to ζ_λ^* . Here the score space is $\mathcal{M}_\tau$ for finite λ , the centered score space for $\lambda = 0$, and $\mathcal{C}_M(\hat{S}_\infty)$ for $\lambda = \infty$.

Lemma D.8 (Verification of GACV regularity). *Suppose the conditions of Lemma D.4 hold, $\mathcal{G}_H^\infty$ is connected, and $\bar{\Lambda}$ is fixed and finite. As $n_L \rightarrow \infty$, Assumption A1 holds with probability tending to one over the LLM sample, simultaneously for $\lambda \in \bar{\Lambda}$.*

Proof of Lemma D.8. For finite $\lambda > 0$, uniform C^2 convergence of ℓ_L to R_L together with Lemmas D.3 and D.4 gives a unique separated regular conditional minimizer, and finite-support M-estimation gives conditional consistency as $n_H \rightarrow \infty$. For $\lambda = 0$, connectivity gives the same properties on the centered score space. For $\lambda = \infty$, Proposition B.5 gives $\widehat{M}_\infty \xrightarrow{P} M$, hence $\text{rank}(D_H^\infty \widehat{M}_\infty) = d$ with probability tending to one, and the fixed-space argument applies. Finiteness of $\bar{\Lambda}$ gives simultaneity. $\square$

Proof of Theorem 5. Condition on the complete LLM comparison sample throughout. For finite $\lambda \in \Lambda$, use Lemma D.1 to identify each regular factorized full-sample fit with its score-level minimizer in $\mathcal{M}_r$; for $\lambda = 0$ use the centered score space, and for $\lambda = \infty$ use $\mathcal{C}_M(\hat{S}_\infty)$. Let ζ denote smooth local coordinates on the neighborhood $\mathcal{U}_\lambda$ in Assumption A1. By that assumption and finiteness of $\bar{\Lambda}$, with conditional probability tending to one all full-sample fits lie in these neighborhoods, their Hessians are uniformly nonsingular, and the required derivatives are uniformly bounded.

For an individual human comparison t , let (I_t, J_t) denote its compared pair and let $Z_t \in \{0, 1\}$ indicate preference for item I_t . Its negative log-likelihood contribution is

$$h_t(s) = \log\{1 + \exp(s_{I_t} - s_{J_t})\} - Z_t(s_{I_t} - s_{J_t}).$$

In local coordinates, write $h_t(\zeta) := h_t\{s(\zeta)\}$. For each candidate λ , the full-sample criterion can be written locally as

$$q_\lambda(\zeta) = \frac{1}{n_H} \sum_{t=1}^{n_H} h_t(\zeta) + a_\lambda(\zeta),$$

where $a_\lambda(\zeta) = \lambda \ell_L\{S(\zeta), b(\zeta)\}$ for finite λ , $a_0 = 0$, and at $\lambda = \infty$ the LLM-only fit is absorbed into the fixed reduced score space. In every case, a_λ is fixed when a human comparison is deleted.

Let $\hat{\zeta}_\lambda$ minimize q_λ and define

$$g_{t,\lambda} := \nabla h_t(\hat{\zeta}_\lambda), \quad \bar{g}_\lambda := \frac{1}{n_H} \sum_{t=1}^{n_H} g_{t,\lambda}, \quad \hat{H}_\lambda := \nabla^2 q_\lambda(\hat{\zeta}_\lambda).$$

The first-order condition gives

$$\bar{g}_\lambda + \nabla a_\lambda(\hat{\zeta}_\lambda) = 0.$$

After deleting human comparison t , we renormalize the human loss by $n_H - 1$ and obtain

$$q_{\lambda,-t}(\zeta) = \frac{1}{n_H - 1} \sum_{u \neq t} h_u(\zeta) + a_\lambda(\zeta).$$

Evaluating its score at the full-sample estimator and using the preceding first-order condition gives

$$\begin{aligned} \nabla q_{\lambda,-t}(\hat{\zeta}_\lambda) &= \frac{n_H \bar{g}_\lambda - g_{t,\lambda}}{n_H - 1} + \nabla a_\lambda(\hat{\zeta}_\lambda) \\ &= -\frac{g_{t,\lambda} - \bar{g}_\lambda}{n_H - 1}. \end{aligned}$$

The individual score vectors are uniformly bounded because the comparison support is finite and the chart derivatives are bounded on the fixed local neighborhoods above, so

$$\max_{\substack{1 \leq t \leq n_H \\ \lambda \in \bar{\Lambda}}} \left\| \nabla q_{\lambda,-t}(\hat{\zeta}_\lambda) \right\|_2 = \mathcal{O}_{\mathbb{P}}(n_H^{-1}).$$

Deleting one human comparison perturbs the criterion by $\mathcal{O}_{\mathbb{P}}(n_H^{-1})$ in C^2 norm on these neighborhoods. By the finite-support uniform law of large numbers and the separation in Assumption A1, with conditional probability tending to one every leave-one-out global minimizer lies in $\mathcal{U}_\lambda$, uniformly over t and $\lambda \in \bar{\Lambda}$. The Hessian remains nonsingular there, and the implicit function theorem gives a unique stationary point near $\hat{\zeta}_\lambda$; hence the leave-one-out global minimizer $\hat{\zeta}_\lambda^{(-t)}$ is this point. A mean-value expansion of the leave-one-out first-order condition, together with uniform Hessian nonsingularity, therefore first gives

$$\max_{\substack{1 \leq t \leq n_H \\ \lambda \in \bar{\Lambda}}} \left\| \hat{\zeta}_\lambda^{(-t)} - \hat{\zeta}_\lambda \right\|_2 = \mathcal{O}_{\mathbb{P}}(n_H^{-1}).$$

At $\hat{\zeta}_\lambda$, deleting one human observation changes the Hessian by $\mathcal{O}_{\mathbb{P}}(n_H^{-1})$ uniformly in t and λ . The preceding perturbation rate and the uniform third-derivative bound further imply that the leave-one-out Hessian at any intermediate

point between $\widehat{\zeta}_\lambda^{(-t)}$ and $\widehat{\zeta}_\lambda$ equals $\widehat{H}_\lambda + \mathcal{O}_\mathbb{P}(n_H^{-1})$ uniformly. Consequently, expanding the leave-one-out first-order condition and using the inverse perturbation formula gives

$$\widehat{\zeta}_\lambda^{(-t)} - \widehat{\zeta}_\lambda = \frac{1}{n_H - 1} \widehat{H}_\lambda^{-1} (g_{t,\lambda} - \bar{g}_\lambda) + \mathcal{O}_\mathbb{P}(n_H^{-2}), \quad (\text{D.12})$$

uniformly over t and $\lambda \in \bar{\Lambda}$.

Because the Hessians of the individual human losses are uniformly bounded on these neighborhoods, (D.12) implies that a first-order Taylor expansion of the deleted human loss has quadratic remainder $\mathcal{O}_\mathbb{P}(n_H^{-2})$ uniformly in t and λ . Thus

$$\begin{aligned} h_t(\widehat{\zeta}_\lambda^{(-t)}) &= h_t(\widehat{\zeta}_\lambda) + g_{t,\lambda}^\top (\widehat{\zeta}_\lambda^{(-t)} - \widehat{\zeta}_\lambda) + \mathcal{O}_\mathbb{P}(n_H^{-2}) \\ &= h_t(\widehat{\zeta}_\lambda) + \frac{1}{n_H - 1} g_{t,\lambda}^\top \widehat{H}_\lambda^{-1} (g_{t,\lambda} - \bar{g}_\lambda) + \mathcal{O}_\mathbb{P}(n_H^{-2}), \end{aligned}$$

uniformly over t and λ . Averaging over t gives

$$\begin{aligned} \text{CV}_{\text{loo}}(\lambda) &= \ell_H(\widehat{s}_\lambda) + \frac{1}{n_H(n_H - 1)} \sum_{t=1}^{n_H} g_{t,\lambda}^\top \widehat{H}_\lambda^{-1} (g_{t,\lambda} - \bar{g}_\lambda) + \mathcal{O}_\mathbb{P}(n_H^{-2}) \\ &= \ell_H(\widehat{s}_\lambda) + \frac{1}{n_H - 1} \text{tr} \left(\widehat{H}_\lambda^{-1} \widehat{J}_\lambda \right) + \mathcal{O}_\mathbb{P}(n_H^{-2}), \end{aligned}$$

because $\sum_t (g_{t,\lambda} - \bar{g}_\lambda) = 0$. This proves (16).

Finally, for finite λ , coordinate invariance of (15) follows from Lemma D.2, which also permits evaluation in the over-parameterized coordinates (D.2). For $\lambda \in \{0, \infty\}$, the same trace identity follows directly under any smooth nonsingular reparameterization of the corresponding score space. $\square$

D.4 Additional results for adaptive calibration

Corollary D.9 (Near-LOO optimality of GACV). *Under the conditions of Theorem 5, conditionally on the LLM comparisons,*

$$\text{CV}_{\text{loo}}(\widehat{\lambda}) \leq \min_{\lambda \in \bar{\Lambda}} \text{CV}_{\text{loo}}(\lambda) + \mathcal{O}_\mathbb{P}(n_H^{-2}). \quad (\text{D.13})$$

Proof of Corollary D.9. Let

$$\Delta_n := \max_{\lambda \in \bar{\Lambda}} |\text{GACV}(\lambda) - \text{CV}_{\text{loo}}(\lambda)| = \mathcal{O}_\mathbb{P}(n_H^{-2})$$

by Theorem 5. If $\lambda_{\text{loo}} \in \arg\min_{\lambda \in \bar{\Lambda}} \text{CV}_{\text{loo}}(\lambda)$, then optimality of $\widehat{\lambda}$ for GACV gives

$$\text{CV}_{\text{loo}}(\widehat{\lambda}) \leq \text{CV}_{\text{loo}}(\lambda_{\text{loo}}) + 2\Delta_n,$$

which proves (D.13). $\square$

For each fixed candidate λ , conditionally on the LLM sample, $\mathbb{E}\{\text{CV}_{\text{loo}}(\lambda) \mid \mathcal{D}_L\} = \mathbb{E}[R_H(\widehat{s}_{\lambda, n_H-1}) \mid \mathcal{D}_L]$, where $\widehat{s}_{\lambda, m}$ denotes the fit using m independent human comparisons. Thus GACV is a second-order approximation to an unbiased estimator of candidate predictive human risk, and GACV achieves near-LOO selection over the fixed grid without a human validation split or a rate restriction between n_H and n_L ; it can likewise select (r, λ) jointly as in Section E.3.1.

Calibration-space test and endpoint comparison. The restriction (4) can be tested using the two endpoint fits already computed for GACV.

Proposition D.10 (Calibration-space likelihood-ratio test). *Suppose that N , K , and r are fixed, that $\mathcal{G}_H^\infty$ is connected, that Proposition B.5 applies, and that $n_H, n_L \rightarrow \infty$ with $n_H/n_L \rightarrow 0$. Define*

$$T_{n_H} := 2n_H \{\ell_H(\widehat{s}_\infty) - \ell_H(\widehat{s}_0)\}. \quad (\text{D.14})$$

Under (4),

$$T_{n_{\text{H}}} \xrightarrow{\text{d}} \chi_{N-1-d}^2.$$

Along (D.10),

$$T_{n_{\text{H}}} \xrightarrow{\text{d}} \chi_{N-1-d}^2(2\delta_M), \quad 2\delta_M = \text{dist}_L^2\{h, \text{col}(M)\}.$$

The AIC comparison between the anchored and unrestricted endpoints selects anchoring exactly when

$$T_{n_{\text{H}}} < 2(N - 1 - d),$$

whose local first-order expected boundary is $\delta_M < (N - 1 - d)/2$, matching Theorem 3.

Proof of Proposition D.10. Write $\widehat{M}_{\infty} := M_{\widehat{\theta}_{\infty}}$. By Proposition B.5, $\|P_{\text{col}(\widehat{M}_{\infty})} - P_{\text{col}(M)}\|_{\text{F}} = \mathcal{O}_{\mathbb{P}}(n_{\text{L}}^{-1/2})$, so $n_{\text{H}}/n_{\text{L}} \rightarrow 0$ and the argument of Corollary C.6 allow $\text{col}(\widehat{M}_{\infty})$ to be replaced by $\text{col}(M)$ with $o_{\mathbb{P}}(1)$ change in $T_{n_{\text{H}}}$. The resulting statistic is the likelihood-ratio test of the d -dimensional restriction $s \in \text{col}(M)$ within the $(N - 1)$ -dimensional centered human BTL model, so Wilks's theorem gives the null limit. Under (D.10), Lemma C.1(iii) gives noncentrality $\text{dist}_L^2\{h, \text{col}(M)\} = 2\delta_M$. Finally, the anchored-minus-unrestricted AIC difference is $T_{n_{\text{H}}} - 2(N - 1 - d)$, and the mean of the local limiting statistic is $N - 1 - d + 2\delta_M$. $\square$

E Experimental and Implementation Details

E.1 DIAL Estimation Procedure

Algorithms E.1 and E.2 summarize the estimation procedures used throughout the experiments. The main reported DIAL estimator uses the consensus calibration basis $M_\theta = \mu$; replacing it by $M_\theta = W_\theta = [\mu, V]$ gives DIAL-Ada (W) without changing the fitting or tuning logic. DIAL-Anc first fits the structured LLM model and then calibrates the resulting representation to the human comparisons, producing the staged LLM-anchored endpoint $(\hat{c}_\infty, \hat{\theta}_\infty, \hat{s}_\infty)$. DIAL-Ada uses this endpoint to initialize a descending path over finite calibration weights and selects among the admissible candidates by GACV. Finite candidate fits that are not numerically resolved within the optimization box, or that fail the numerical regularity checks in Section E.3.2, are excluded from GACV selection. The staged endpoint and the human-only endpoint are included whenever they satisfy the corresponding endpoint regularity checks, with the staged endpoint providing the computational fallback whenever admissible. Replications with no admissible candidate are reported as failures. For GACV, $\hat{H}_\lambda^{-1}$ is evaluated in a nonsingular reduced chart; equivalently, the Moore–Penrose inverse $\hat{H}_{\lambda, \varphi}^+$ may be used in the over-parameterized coordinates of (D.2), by Lemma D.2. Diagnostics for position bias are always reported from the LLM-only component $\hat{\theta}_\infty$, because finite- λ joint fitting allows the human likelihood to perturb the LLM parameters under misspecification. For numerical robustness, optimization is performed over a large box with coordinate bounds 10 in the centering-reduced chart.

Algorithm E.1 DIAL-Anc estimation

Require: LLM cell counts $\{n_{kij}^{(a)}, Y_{kij}^{(a)}\}$; human comparisons $\{n_{Hij}, Y_{Hij}\}$; rank r

Ensure: DIAL-Anc score $\hat{s}_{\text{Anc}} = \hat{s}_\infty$;

1: Set the calibration basis

$$M_\theta \leftarrow \mu, \quad \text{or } M_\theta \leftarrow W_\theta = [\mu, V] \text{ for full-space calibration.}$$

2: Fit the structured order-effect LLM model

$$\hat{\theta}_\infty \in \operatorname{argmin}_{\theta \in \Theta_r} \ell_L(S_\theta, b_\theta).$$

3: Fit the staged human calibration

$$\hat{c}_\infty \in \operatorname{argmin}_c \ell_H(M_{\hat{\theta}_\infty} c).$$

4: Set

$$\hat{s}_\infty = \hat{s}_{\text{Anc}} = M_{\hat{\theta}_\infty} \hat{c}_\infty.$$

▷ Consensus calibration when $M_\theta = \mu$

E.2 Variance Estimation for the DIAL Intervals

All DIAL intervals in this paper are Wald intervals from the sandwich (D.8), whose bread $\hat{\mathcal{H}}_\lambda$ is the Hessian of the weighted criterion and whose meat $\hat{\mathcal{J}}_\lambda$ estimates the variance of its score. Dependence between comparisons changes only the meat, which we estimate in the heteroskedasticity- and autocorrelation-consistent (HAC) form of Newey & West (1987) and Andrews (1991). Write $\mathcal{O}$ for the set of all $n_H + n_L$ comparisons and let $\tilde{\psi}_v := n_H^{-1}(g_{Ht} - \bar{g}_H)$ if v is the human comparison t and $\tilde{\psi}_v := \lambda n_L^{-1}(g_{Lu} - \bar{g}_L)$ if v is the LLM comparison u , the centered contribution of v to the score of q_λ at $\hat{\zeta}_\lambda$. For a symmetric matrix $\kappa = (\kappa_{vw})_{v,w \in \mathcal{O}}$ of kernel weights with unit diagonal, set

$$\hat{\mathcal{J}}_\lambda^\kappa := n_H \sum_{v,w \in \mathcal{O}} \kappa_{vw} \tilde{\psi}_v \tilde{\psi}_w^\top, \tag{E.1}$$

and use it in place of $\hat{\mathcal{J}}_\lambda$ in (D.8); the result is positive semidefinite whenever κ is. The identity kernel returns (D.8), which treats every comparison as independent.

Algorithm E.2 DIAL-Ada estimation

Require: LLM cell counts $\{n_{kij}^{(a)}, Y_{kij}^{(a)}\}$; human comparisons $\{n_{Hij}, Y_{Hij}\}$; rank r ; candidate weights $\bar{\Lambda}$ containing 0 and ∞

Ensure: DIAL-Ada score $\hat{s}_{\text{Ada}}$ and selected weight $\hat{\lambda}$

- 1: Initialize the admissible set $\mathcal{R} \leftarrow \emptyset$.
 - 2: **if** the staged endpoint $(\hat{c}_\infty, \hat{\theta}_\infty, \hat{s}_\infty)$ from Algorithm E.1, Lines 2–4, passes the endpoint regularity check **then**
 - 3: $\mathcal{R} \leftarrow \mathcal{R} \cup \{\infty\}$.
 - 4: **end if**
 - 5: Fit the centered human-only BTL estimator $\hat{s}_0$.
 - 6: **if** $\hat{s}_0$ passes the endpoint regularity check **then**
 - 7: $\mathcal{R} \leftarrow \mathcal{R} \cup \{0\}$.
 - 8: **end if**
 - 9: Order $\Lambda = \bar{\Lambda} \setminus \{0, \infty\}$ from largest to smallest and initialize the path at $(\hat{c}_\infty, \hat{\theta}_\infty)$ from Algorithm E.1, Lines 2–3.
 - 10: **for** $\lambda \in \Lambda$ in descending order **do**
 - 11: Fit

$$(\hat{c}_\lambda, \hat{\theta}_\lambda) \in \underset{c, \theta \in \Theta_r}{\operatorname{argmin}} \{ \ell_H(M_\theta c) + \lambda \ell_L(S_\theta, b_\theta) \},$$
 using the most recent retained solution as a warm start.
 - 12: Set

$$\hat{s}_\lambda = M_{\hat{\theta}_\lambda} \hat{c}_\lambda.$$
 - 13: **if** the fit passes the numerical and regularity checks **then**
 - 14: $\mathcal{R} \leftarrow \mathcal{R} \cup \{\lambda\}$.
 - 15: Retain the fit to warm-start the next weight.
 - 16: **end if**
 - 17: **end for**
 - 18: **if** $\mathcal{R} = \emptyset$ **then**
 - 19: Report the replication as a numerical failure.
 - 20: **else**
 - 21: Compute $\text{GACV}(\lambda)$ for all $\lambda \in \mathcal{R}$.
 - 22: Select

$$\hat{\lambda} \in \underset{\lambda \in \mathcal{R}}{\operatorname{argmin}} \text{GACV}(\lambda), \quad \hat{s}_{\text{Ada}} = \hat{s}_{\hat{\lambda}}.$$
 - 23: **end if**
-

The kernel we use. The binomial model (1) treats the comparisons in a cell (k, i, j) as independent draws with a common probability, whereas the probability may vary with the pair beyond what the scores predict, so that repeated queries of one pair by one judge are correlated through a shared pair-level effect. Our implementation therefore sets $\kappa_{vw} = 1$ when v and w are LLM comparisons of the same judge on the same pair, in either display order, and $\kappa_{vw} = \mathbb{1}\{v = w\}$ otherwise, the cluster-robust form of Liang & Zeger (1986) with judge-pair cells as clusters. This gives

$$\hat{\mathcal{J}}_\lambda^{\text{cell}} = \hat{\mathcal{V}}_H + \lambda^2 \frac{n_H}{n_L} \hat{\mathcal{V}}_L^{\text{cell}}, \quad \hat{\mathcal{V}}_L^{\text{cell}} := \frac{1}{n_L} \sum_{k=1}^K \sum_{i < j} G_{kij} G_{kij}^\top, \quad (\text{E.2})$$

where $G_{kij} := \sum_{a \in \{-1, 1\}} \{n_{kij}^{(a)} \hat{p}_{kij}^{(a)} - Y_{kij}^{(a)}\} \nabla_\zeta \eta_{kij}^{(a)}(\hat{\zeta}_\lambda) - (n_{kij}^{(1)} + n_{kij}^{(-1)}) \bar{g}_L$ is the summed centered score of the cell and $\hat{p}_{kij}^{(a)} := \sigma\{\eta_{kij}^{(a)}(\hat{\zeta}_\lambda)\}$, so (E.2) is computed from the cell counts alone. It matches the dependence generated in the simulations of Section 6.1, where the human comparisons are independent, so only the LLM block is clustered, and it is used for every DIAL interval in the paper except the binomial comparison in Section F. The judge diagnostic of Figure G5(a) reports an LLM-side parameter and uses the same meat in its LLM-only specialization, $\hat{\mathcal{V}}_L^{\text{cell}}$ with the Hessian of ℓ_L/n_L as bread, which is the $\lambda \rightarrow \infty$ limit of (E.2). On the real data a record additionally supplies the judgments of all judges and at most one human label on the same two responses; the kernel that joins all comparisons of a record adds a cross term between the human and LLM scores of each record, requires record identities that the aggregated cell counts discard, and is not used here. Consistency of (E.1) requires the dependence to lie within the

support of κ and the number of independent blocks to grow (Hansen & Lee, 2019); with N and K fixed the cell kernel averages over $K \binom{N}{2}$ cells, and under the pair-level effect it estimates the variance over both the binomial sampling and the draw of the cell effects, which is why it stays near nominal in Section F while the binomial version under-covers as n_L grows.

E.3 Hyperparameter Selection

E.3.1 Rank r

Theorem C.4 applies to any fixed centered subspace, in particular to the nested family $\text{col}(W_{r'})$, $r' = 0, 1, \dots$, obtained by retaining the leading r' disagreement directions of the fitted score matrix. Along this family the approximation error $\Delta_{W_{r'}}$ is nonincreasing in r' , while the first-order estimation term grows as $(r' + 1)/(2n_H)$, so the population rule is to choose the r' minimizing $\delta_{W_{r'}} + (r' + 1)/2$. We implement its empirical counterpart by evaluating (15) over a grid of (r, λ) pairs and selecting the pair with the smallest criterion value, which requires no separate rank-selection procedure.

E.3.2 Weight λ

For each experiment, $\bar{\Lambda}$ is a finite logarithmic grid augmented with the endpoints 0 and ∞ , and $\hat{\lambda}$ is chosen by (15). Theorem 5 treats this candidate set as fixed asymptotically; in the implementation we center the finite grid at the joint-likelihood weight n_L/n_H , using seven log-spaced multiples between 0.01 and 10 (between 0.1 and 10 in the simulations), so that the ordinary joint MLE is always a candidate; the lower end matters on real data, where n_L/n_H is in the hundreds and the path leaves the staged fit only below $0.1 n_L/n_H$. The regularity in Assumption A1 is verified asymptotically by Lemma D.8; in finite samples, a candidate fit can be nonfinite or have a singular criterion Hessian. In particular, the human-only endpoint has no finite MLE when the human comparison graph is disconnected or separated, and the trace term in (15) is then undefined. We therefore drop a finite positive- λ candidate from the GACV argmin when its fit does not converge, is not numerically resolved within the optimization box, has more zero Hessian eigenvalues than the $r(r + 1)$ exact invariances of the computational parameterization in Section D.1, or has $\|\hat{s}_\lambda\|_\infty > 10$. The human-only endpoint is retained only when its directed win graph is strongly connected (Ford, 1957); at $\lambda = \infty$, the LLM-only fit is conditioned on, so only its calibration coordinates must be numerically resolved. The implementation uses $\lambda = \infty$ as the computational fallback; if no candidate passes the guards, that replication is reported as a failure rather than averaged. GACV at the endpoints uses the same formula with the chart of the corresponding estimator, the centered score space at $\lambda = 0$ and the coordinate c with W fixed at the LLM-only fit at $\lambda = \infty$. The path is fitted from the largest λ downward, initializing the first fit at the staged estimator and each subsequent fit at the previous one, and every fit is finished by a quasi-Newton step on the reduced chart.

F Simulation Studies

This section gives the design and implementation details for Figure 2 and three additional synthetic configurations.

F.1 Design and implementation

Data-generating process. We draw the consensus direction μ from a Gaussian distribution, center it, and scale it to $\|\mu\|_2^2 = N$. The columns of $V \in \mathbb{R}^{N \times r}$ are Gaussian, projected orthogonally to 1_N and μ , and normalized so that $N^{-1}V^\top V = I_r$. We take $\gamma_k = 1 + 0.3z_k$ with Gaussian z_k and rescale γ to $1_K^\top \gamma = K$, while U is Gaussian with zero column means, giving $S = \gamma\mu^\top + UV^\top$ satisfying Condition 3. Position effects are $b_k = \beta_k \xi_k$, where $\beta_k \sim \text{Uniform}(0.3, 1.2)$ and $\Pr(\xi_k = 1) = 0.8$, with one judge fixed at $b_k = 0.05$. The main configuration uses $s_H = \mu$, while the misaligned configuration in Figure F2 uses $s_H = Wc_H$ with $c_H = (1, c_V)$ and $c_V \sim \mathcal{N}(0, 0.5^2 I_r)$. Each comparison samples its item pair uniformly, LLM comparisons additionally sample the judge uniformly, and the canonical item is displayed first with probability 0.75. For the bottom-row experiments, we replace the LLM linear predictor by $\eta_{kij}^{(a)} + \varepsilon_{kij}$ with $\varepsilon_{kij} \sim \mathcal{N}(0, 1)$ drawn once per judge-pair and shared across repeated queries and both display orders. For Figure F3, we instead generate pair-varying position effects $b_{kij} = b_k + \delta_{kij}$ and use the linear predictor $s_{ki} - s_{kj} + ab_{kij}$, where δ_{kij} is drawn once per judge-pair, centered within judge over pairs, and has pairwise standard deviation σ_{pos} ; estimation retains the working model with a single b_k per judge.

Configurations. Figure 2 uses $(N, K, r) = (10, 4, 1)$, with $n_L = 20,000$ and $n_H \in \{100, 200, 400, 800, 1600\}$ in the top row, and $n_H = 800$ with $n_L \in \{400, 800, \dots, 12,800\}$ in the bottom row. Figure F1 uses $(N, K, r) = (20, 6, 2)$, with $n_L = 60,000$ and $n_H \in \{300, \dots, 4800\}$ in the top row, and $n_H = 2400$ with $n_L \in \{2400, \dots, 76,800\}$ in the bottom row. Figure F2 uses the same budgets as Figure 2 with the misaligned human target above. Figure F3 uses the main $(N, K, r) = (10, 4, 1)$ configuration with $s_H = \mu$, $n_H = 800$, $n_L = 20,000$, and $\sigma_{\text{pos}} \in \{0, 0.25, 0.5, 0.75, 1\}$. Every configuration uses 50 independent replications.

Estimation and metrics. All structured estimators use the true working rank r and are defined as in Section 6; finite- λ paths are warm-started from the staged endpoint and use the GACV grid and regularity checks of Section E.3.2. We additionally evaluate the oracle candidate minimizing population human risk. The reported metrics are excess human risk $R_H(\hat{s}) - R_H(s_H)$, Kendall’s τ between $\hat{s}$ and s_H , $\text{RMSE}(\hat{b}, b)$, and coverage of pairwise human-score contrasts. Coverage uses Fisher information for Human and the judge-pair clustered sandwich of Section E.2 for DIAL estimators; intervals at the GACV-selected weight are post-selection. Since Pooled and DIAL-noPos impose $\hat{b} = 0$, their order-effect errors coincide, with Pooled drawn as a wider band beneath DIAL-noPos in the corresponding panels.

Numerical summary of Figure 2. Under exact specification (top row), Human tracks the $(N - 1)/(2n_H)$ first-order risk benchmark of Theorem 3 and DIAL-Anc the $1/(2n_H)$ benchmark of its one-dimensional calibration space; at $n_H \leq 200$, DIAL-Anc reduces excess risk by 8–14 $\times$ and raises Kendall’s τ from 0.75 to 0.98 relative to Human. The median excess risk of DIAL-Ada stays within 20% of DIAL-Anc, whereas DIAL-Ada(W) is 1.3–1.6 $\times$ higher because it estimates an unused disagreement direction. Under pair-level LLM misspecification (bottom row), DIAL-Anc plateaus and is worse than Human for $n_L \leq 6400$, while DIAL-Ada is below both endpoints at every n_L and within 0.001 of the oracle; its Kendall’s τ is 0.91–0.93, against 0.89 for Human and 0.78–0.90 for DIAL-Anc. For order-aware fits, $\text{RMSE}(\hat{b}, b)$ falls from 0.37 to 0.11 across the bottom row and reaches 0.039 at $n_L = 20,000$, while methods imposing $\hat{b} = 0$ remain at 0.68, and cell-clustered coverage is 0.93–0.96.

F.2 Additional configurations

Larger problem. Figure F1 repeats the experiment at $(N, K, r) = (20, 6, 2)$. The main patterns persist: under exact specification DIAL-Ada approaches DIAL-Anc as human supervision becomes scarce, while under LLM overdispersion it remains below both Human and DIAL-Anc across the LLM-budget sweep. Judge-pair clustered coverage remains between 0.92 and 0.96, and DIAL-Ada(W) pays additional variance under the consensus-aligned target because its extra disagreement directions are unnecessary.

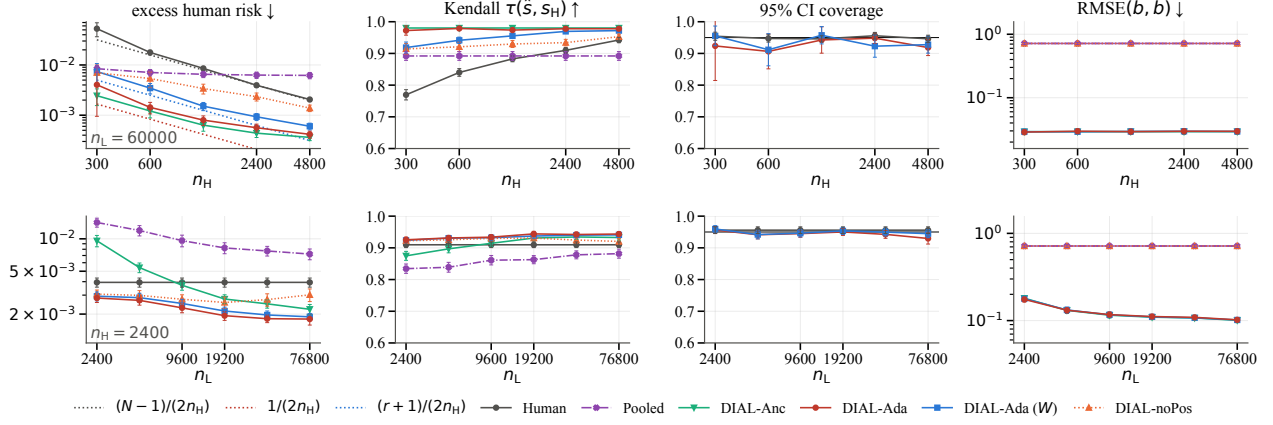


Figure F1: Larger-scale replicate of Figure 2 with $N = 20$, $K = 6$, $r = 2$, and 50 replications; rows, columns, and estimators follow Figure 2.

Human target outside the consensus direction. Figure F2 sets $s_H = Wc_H$ with $c_V \sim \mathcal{N}(0, 0.5^2)$, so calibration within $\text{col}(W)$ is correctly specified but consensus-only calibration has nonzero approximation error Δ_μ . DIAL-Anc and Pooled plateau at excess risks of about 0.040 and 0.045, while DIAL-Ada shifts toward the human comparisons as n_H grows but cannot represent the missing disagreement direction. In contrast, DIAL-Ada(W) continues to decrease with the human budget, attaining Kendall's τ of 0.93–0.97 versus 0.83–0.94 for DIAL-Ada and coverage of 0.94–0.96 under the correctly specified top-row model. Under bottom-row LLM overdispersion, DIAL-Ada, DIAL-Ada(W), and Human are within 0.001 in excess risk, while DIAL-Anc remains biased.

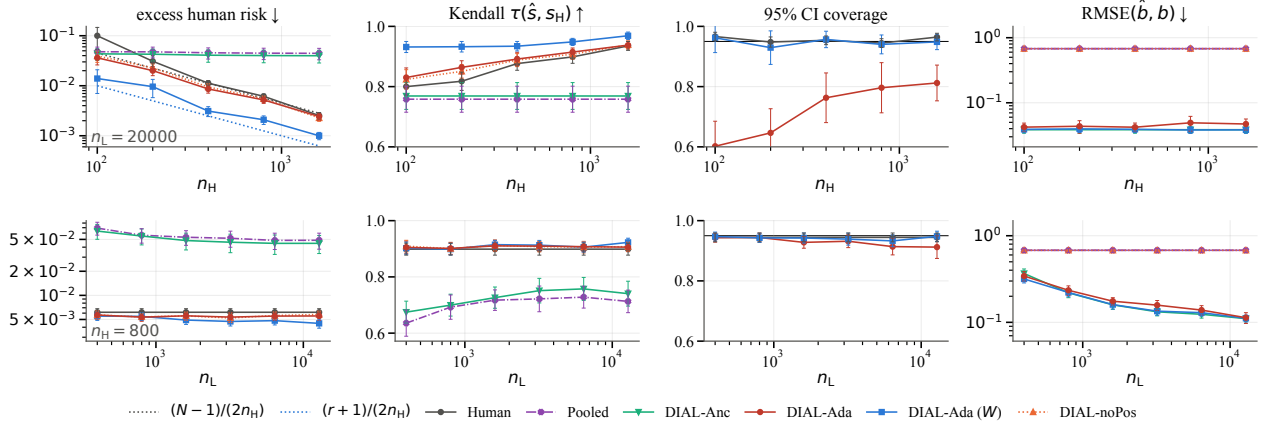


Figure F2: Misaligned-target simulation under the main $(N, K, r) = (10, 4, 1)$ configuration, with $s_H = Wc_H$ and $c_V \sim \mathcal{N}(0, 0.5^2)$. Rows and columns follow Figure 2; dotted lines show the first-order excess-risk benchmarks for Human and calibration within $\text{col}(W)$.

Pair-varying position effects. Figure F3 violates the constant-position-effect working model by replacing b_k with $b_k + \delta_{kij}$, while keeping the latent LLM score structure and human calibration model correctly specified. As σ_{pos} increases from 0 to 1, the excess human risk of DIAL-Anc and DIAL-Ada rises from 0.0011 and 0.0013 to 0.0030 and 0.0027 and their Kendall's τ falls from 0.978 and 0.972 to 0.932 and 0.928, yet both stay better than Human (0.0059, $\tau = 0.894$) and DIAL-noPos at every σ_{pos} . The fitted $\hat{b}_k$ summarizes the misspecified constant-effect component and is evaluated against the judge-level mean b_k ; so its RMSE grows from 0.039 to 0.114 while remaining about six times smaller than the 0.679 of the order-agnostic Pooled and DIAL-noPos, which impose $\hat{b} = 0$.

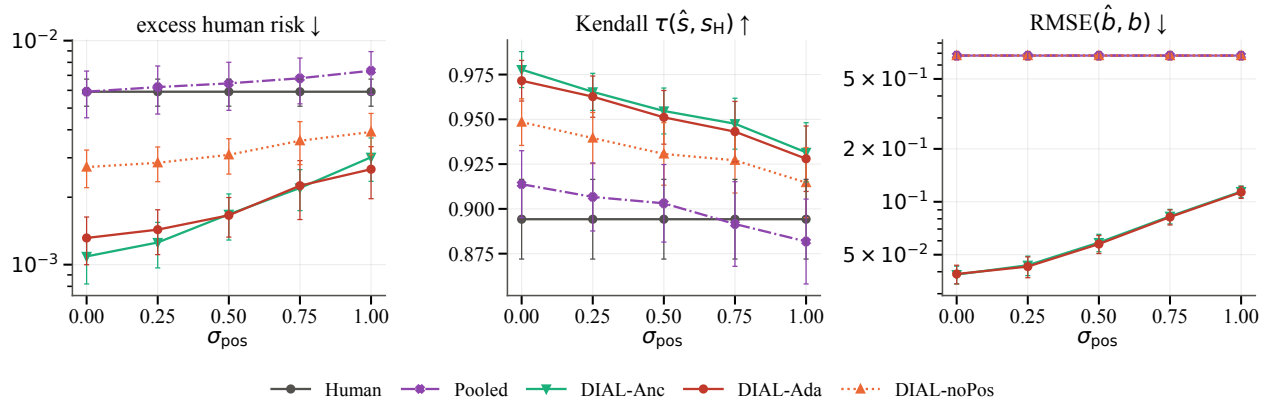


Figure F3: Robustness to heterogeneous position effects under the main $(N, K, r) = (10, 4, 1)$ configuration with $n_H = 800$ and $n_L = 20,000$. Data are generated with $b_{kij} = b_k + \delta_{kij}$, where the within-judge pair effects have standard deviation σ_{pos} , while all methods retain a constant b_k working model. Panels report excess human risk, Kendall's τ , and $\text{RMSE}(\hat{b}, b)$ over 50 replications.

G Real Data

G.1 Datasets and collection

We use Chatbot Arena (Chiang et al., 2024), MT-Bench (Zheng et al., 2023), and PandaLM (Wang et al., 2024b), whose records contain a prompt, two responses, and a human preference label. For every record we queried 22 LLM judges in both display orders: 19 open-weight models via Ollama and 3 commercial APIs (Claude Haiku 4.5, Gemini 3.1 Flash Lite, and GPT-4.1 nano); decoding configurations are provided with the released collection kit. We exclude StableLM 2, 12B because its tie rate exceeds 50% on every dataset, leaving $K = 21$ judges.

Because (1) and (3) are binary, we exclude LLM ties and human ties or “both bad” labels; treating ties as half wins produced the same qualitative conclusions. Because asymmetric tie rates can also carry position information, explicit tie modeling remains an extension. Table G1 reports the retained and discarded counts.

Table G1: Dataset statistics after preprocessing. “Kept” refers to the 21 retained judges; a cell is a (judge, canonical pair) combination.

	Chatbot Arena	MT-Bench	PandaLM
Records (prompts with a human label)	7,738	1,199	945
Candidate models N	20	6	5
Observed pairs / possible	189 / 190	15 / 15	10 / 10
Human labels a / b / tie / both bad	2,771 / 2,659 / 814 / 1,494	432 / 410 / 357 / 0	398 / 444 / 103 / 0
Decisive human labels	5,430	842	842
LLM judgments, kept judges	321,668	49,172	39,324
of which ties	21,199 (6.6%)	3,289 (6.7%)	4,374 (11.1%)
tie-excluded n_L	300,469	45,883	34,950
Display order +1 / -1	150,299 / 150,170	22,909 / 22,974	17,443 / 17,507
Cells observed / possible	3,965 / 3,990	315 / 315	210 / 210
n_{kij} per cell (min / mean / max)	1 / 75.8 / 381	64 / 145.7 / 215	114 / 166.4 / 208
Judge tie rate (min-max)	0.002-0.175	0.004-0.145	0.004-0.247

Table G2: Candidate models treated as items after preprocessing.

Dataset	Candidate models
Chatbot Arena ($N = 20$)	RWKV-4-Raven-14B, alpaca-13b, chatglm-6b, claude-instant-v1, claude-v1, dolly-v2-12b, fastchat-t5-3b, gpt-3.5-turbo, gpt-4, gpt4all-13b-snoozy, guanaco-33b, koala-13b, llama-13b, mpt-7b-chat, oasst-pythia-12b, palm-2, stablelm-tuned-alpha-7b, vicuna-13b, vicuna-7b, wizardlm-13b
MT-Bench ($N = 6$)	alpaca-13b, claude-v1, gpt-3.5-turbo, gpt-4, llama-13b, vicuna-13b-v1.2
PandaLM ($N = 5$)	bloom-7b, cerebras-gpt-6.7b, llama-7b, opt-7b, pythia-6.9b

Chatbot Arena is the largest and sparsest benchmark, whereas MT-Bench and PandaLM have complete pair coverage but only 6 and 5 candidate models (Table G2); PandaLM’s original display order is one-sided. Each record contributes up to 42 LLM judgments but at most one human label, motivating adaptive weighting when within-record judgments are dependent.

G.2 Protocol and estimators

We split records into train/test pools (70/30% for Chatbot Arena and 50/50% otherwise), keeping both display orders of each record together. Human calibration samples are drawn from decisive train labels, while LLM samples use tie-excluded train judgments subject to the specified judge panel and display design. We report excess held-out human log loss relative to a test-pool BTL fit and Kendall’s τ against its ranking. All curves show means over up to 50 record splits with ± 1.96 Monte Carlo standard errors.

We use the estimators in Section 6 with $r = 1$; DIAL-Ada (W) is retained in Figure 3 but omitted from the remaining curve figures for readability. Additional fixed-weight, oracle-weight, and rank-selection diagnostics use the same GACV grid and finite-fit rules as Section E.3.2. Judges with no retained judgments after subsampling are removed for that replication. A bound-doubling sensitivity check gives essentially unchanged results except in the

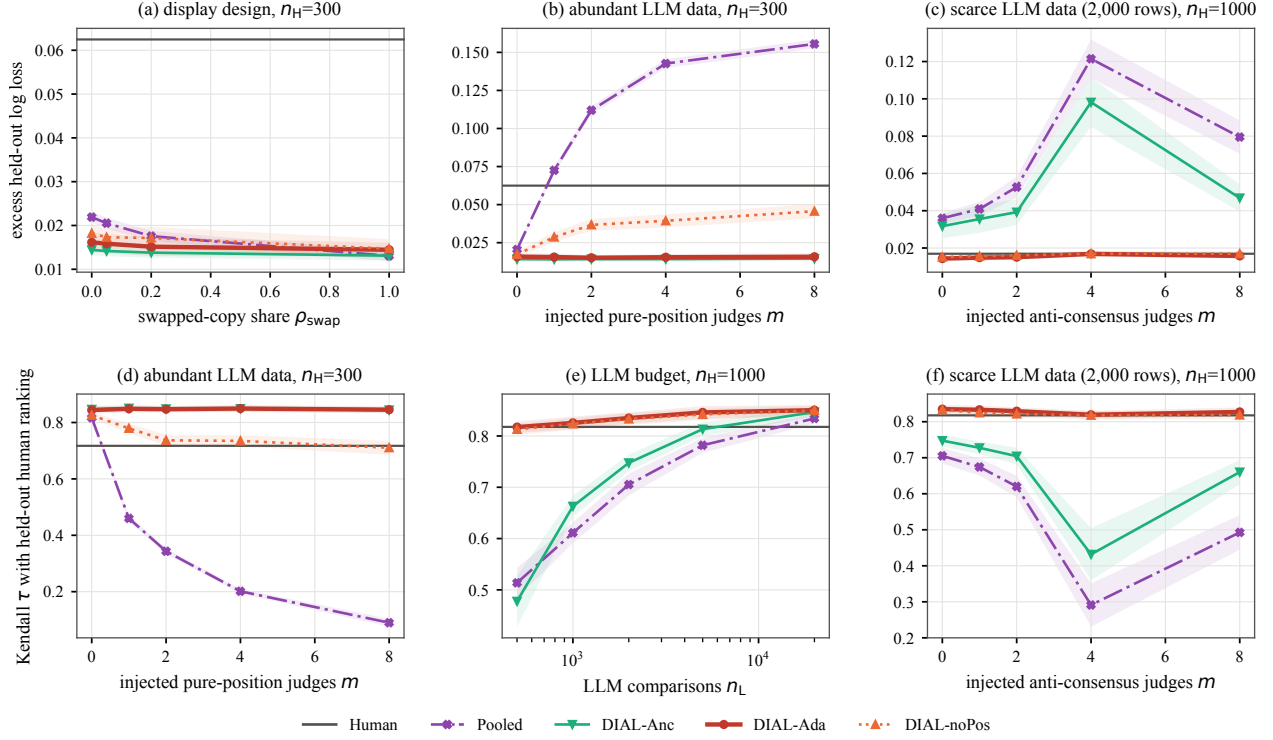


Figure G4: Real-data robustness sweeps on Chatbot Arena using five order-sensitive judges. Top: excess held-out log loss under (a) display imbalance, (b) injected pure-position judges, and (c) injected anti-consensus judges with scarce LLM data. Bottom: Kendall’s τ under (d) the pure-position perturbation, (e) a balanced LLM-budget sweep, and (f) the anti-consensus perturbation. Means over up to 50 record splits with ± 1.96 Monte Carlo standard errors.

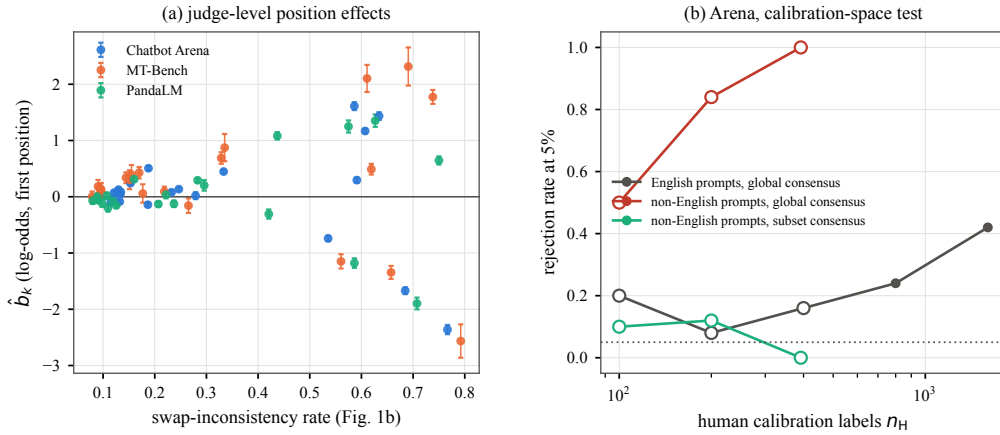


Figure G5: Additional real-data diagnostics. (a) LLM-only fitted order effects of the 21 judges, with cell-clustered 95% intervals, against swap-inconsistency rate. (b) Rejection rate of the calibration-space likelihood-ratio test of $s_H = \alpha\mu$ on Chatbot Arena; hollow markers denote budgets at which the unrestricted human-only MLE failed in some replications, and at the two smallest budgets it failed in most, so those points are not interpreted.

sparsest LLM regimes, where the LLM-only fit is numerically unresolved at either bound; at the smallest LLM budgets and judge panels we therefore read the curves as trends rather than as exact magnitudes or orderings. Replications in which no candidate passes the guards of Section E.3.2 have no admissible fit and are reported as failures rather than averaged, which affects at most three of the 50 splits in any reported cell.

G.3 Position robustness evaluations and diagnostics

Display imbalance and pure-position judges. The five-judge panel contains the judges with the largest swap-inconsistency on Chatbot Arena and is fixed across datasets, with $n_H = 300, 80, 60$. We retain every canonical-first judgment and a fraction ρ_{swap} of swapped copies, with $\rho_{\text{swap}} = 1$ balanced and $\rho_{\text{swap}} = 0.05$ used for the one-sided design in Figure 3; Figure G4(a) sweeps ρ_{swap} to zero. For the injected perturbation, each pure-position judge selects the first-displayed response with probability 0.9, independent of item identity. Order-aware DIAL remains stable as display imbalance increases and pure-position judges are added, whereas the order-agnostic baselines deteriorate (Figures 3 and G4(a,b,d)).

Robustness to imperfect LLM consensus. For the LLM-budget experiments, the balanced five-judge panel is subsampled to n_L rows at fixed $n_H = 1000, 80, 60$ on Chatbot Arena, MT-Bench, and PandaLM, respectively. Injected anti-consensus judges reproduce the flipped verdicts of a randomly selected panel judge; Figure G4(c,f) uses the scarce-LLM setting. Under this corruption, DIAL-Ada shifts toward the human evidence and is substantially more stable than fixed anchoring, although behavior at the strongest corruption is non-monotone.

Judge-specific position effects. Figure G5(a) compares the LLM-only $\hat{b}_k$ from the full balanced data with each judge’s swap-inconsistency rate, using cell-clustered intervals from Section E.2. The fitted effects vary substantially across judges in both magnitude and sign and broadly track empirical order sensitivity.

G.4 Human-label efficiency and adaptive calibration

Calibration-space diagnostic. Chatbot Arena records are split by recorded prompt language (6,910 English, 828 other), with 30% of each subset held out for testing. We apply Proposition D.10 at the 5% level using either the global rank-one consensus or a consensus refitted within the subset. The statistic is defined only where the unrestricted human BTL fit is finite, so rejection rates are read at the budgets where that fit exists in all but a few splits: at the full non-English calibration pool it is finite in 48 of the 50 splits, and restricting to those splits leaves the rejection rates unchanged at 1.00 for the global consensus and 0.00 for the refitted one, whereas at 100 and 200 labels it is finite in only 4% and 38% of splits and those budgets are not interpreted. Figure G5(b) shows a pronounced mismatch of the global consensus for non-English prompts that largely disappears after subset-specific refitting, which also improves held-out ranking and log loss.

Human-label efficiency with abundant LLM data. With all 21 judges, DIAL-Anc and DIAL-Ada remain strong at small human-label budgets, where Human often has no finite MLE (Figure G6, top). When LLM evidence is abundant and balanced, adaptive updating offers little gain over anchoring; across ranks, the oracle calibration gains are negligible, providing little evidence that disagreement directions improve the pooled human target.

Adaptive calibration with scarce LLM data. When the five-judge panel is subsampled to small n_L , the LLM-only consensus becomes noisy and DIAL-Anc degrades, whereas DIAL-Ada uses the human comparisons to update the consensus and remains competitive with the better endpoint across datasets (Figure G6, bottom). As n_L grows, the selected fit moves toward anchoring and the estimators converge.

Judge-panel and budget sensitivity. Figure G7 compares six-judge open-weight panels with the full 21-judge panel across human-label budgets. The small panel remains close to the larger panels on Chatbot Arena and PandaLM, with MT-Bench the main exception; at the smallest human budgets, DIAL-Ada is less stable than DIAL-Anc on the small panel. Thus the human-label gains of DIAL do not require the full judge panel, although weaker panels leave less margin for adaptive updating.

Adaptive calibration across LLM budgets. With abundant LLM judgments and limited human supervision, DIAL-Anc is competitive because the LLM consensus is well estimated. As the LLM budget decreases, DIAL-Ada increasingly improves by using the human comparisons to update the noisier consensus estimate (Figure G8). This pattern holds across judge panels and complements the human-budget sweep in Figure G7.

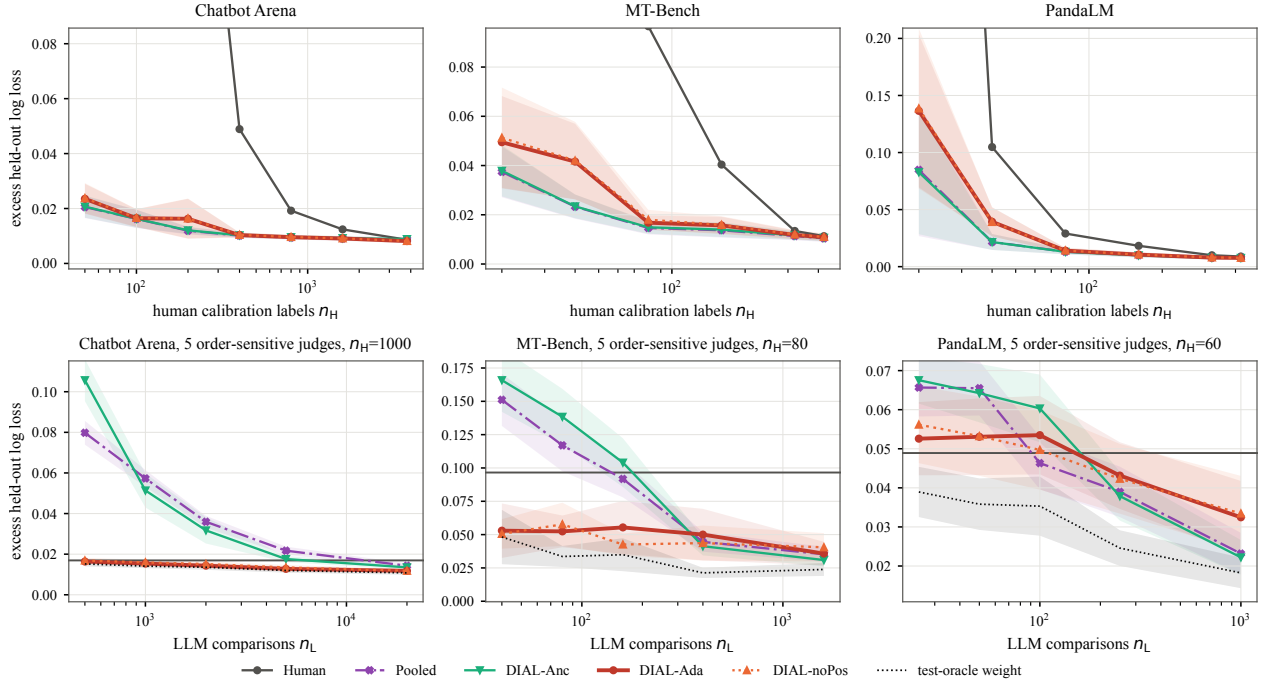


Figure G6: Human-label efficiency and LLM-budget sensitivity. Top: excess held-out human log loss against the human budget n_H with the as-collected balanced judgments of all 21 judges; Human is off the chart at the smallest budgets, where its MLE does not exist. Bottom: the five order-sensitive judges' balanced judgments subsampled to n_L rows at a fixed calibration sample ($n_H = 1,000, 80, 60$); the dotted line is the candidate weight minimizing the test log loss. Means over up to 50 record splits with ± 1.96 Monte Carlo standard errors.

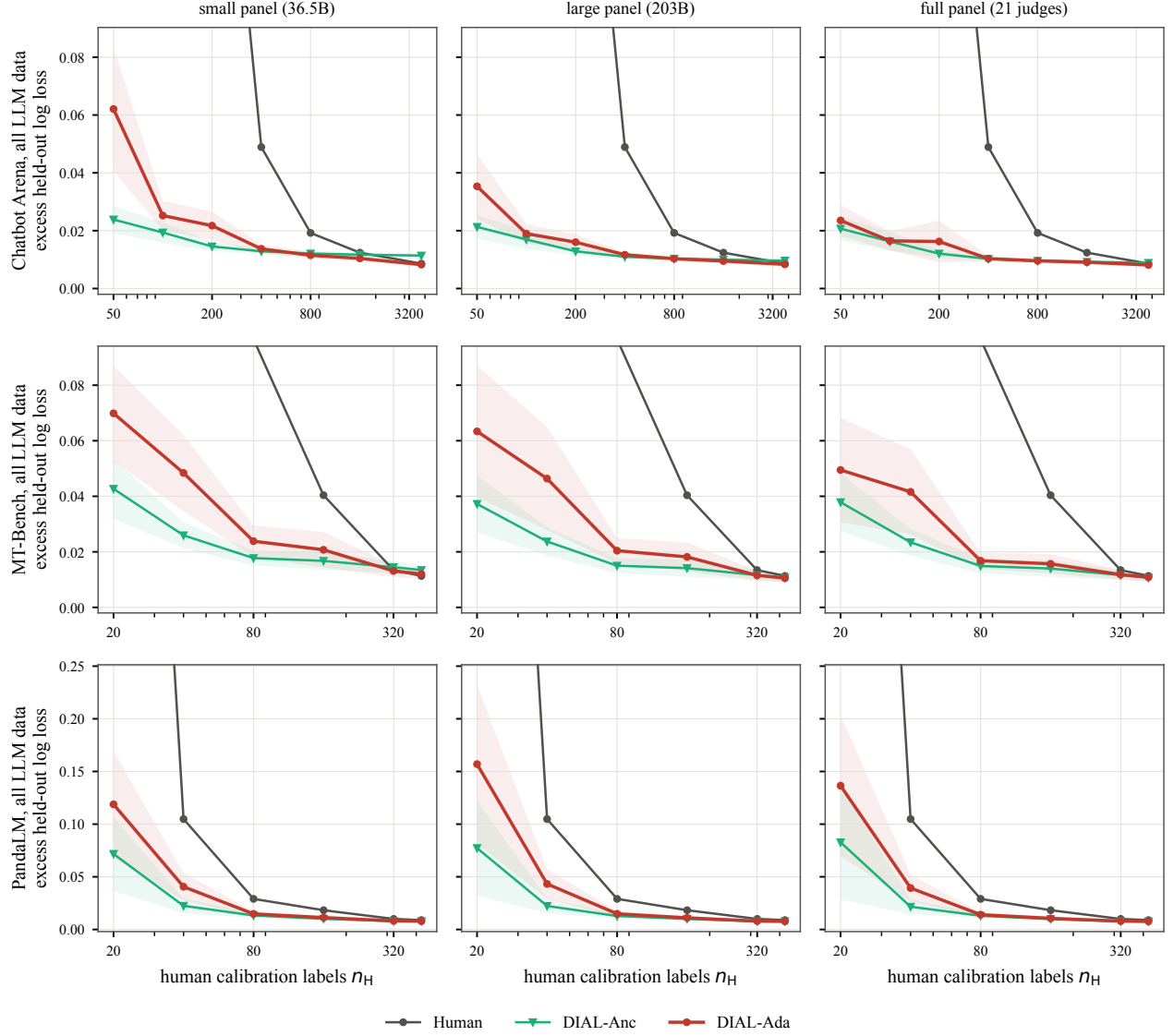


Figure G7: Excess held-out human log loss against the human budget n_H , as in Figure G6 (top), on the small, large, and full judge panels (columns) across Chatbot Arena, MT-Bench, and PandaLM (rows). The small and large panels each contain six open-weight judges with disclosed parameter counts $\leq 9B$ and $\geq 20B$ per judge, respectively (36.5B versus 203B total); full is the 21-judge panel. All curves use up to the same 50 record splits, with ± 1.96 Monte Carlo standard errors; the y-axis is shared within each row.

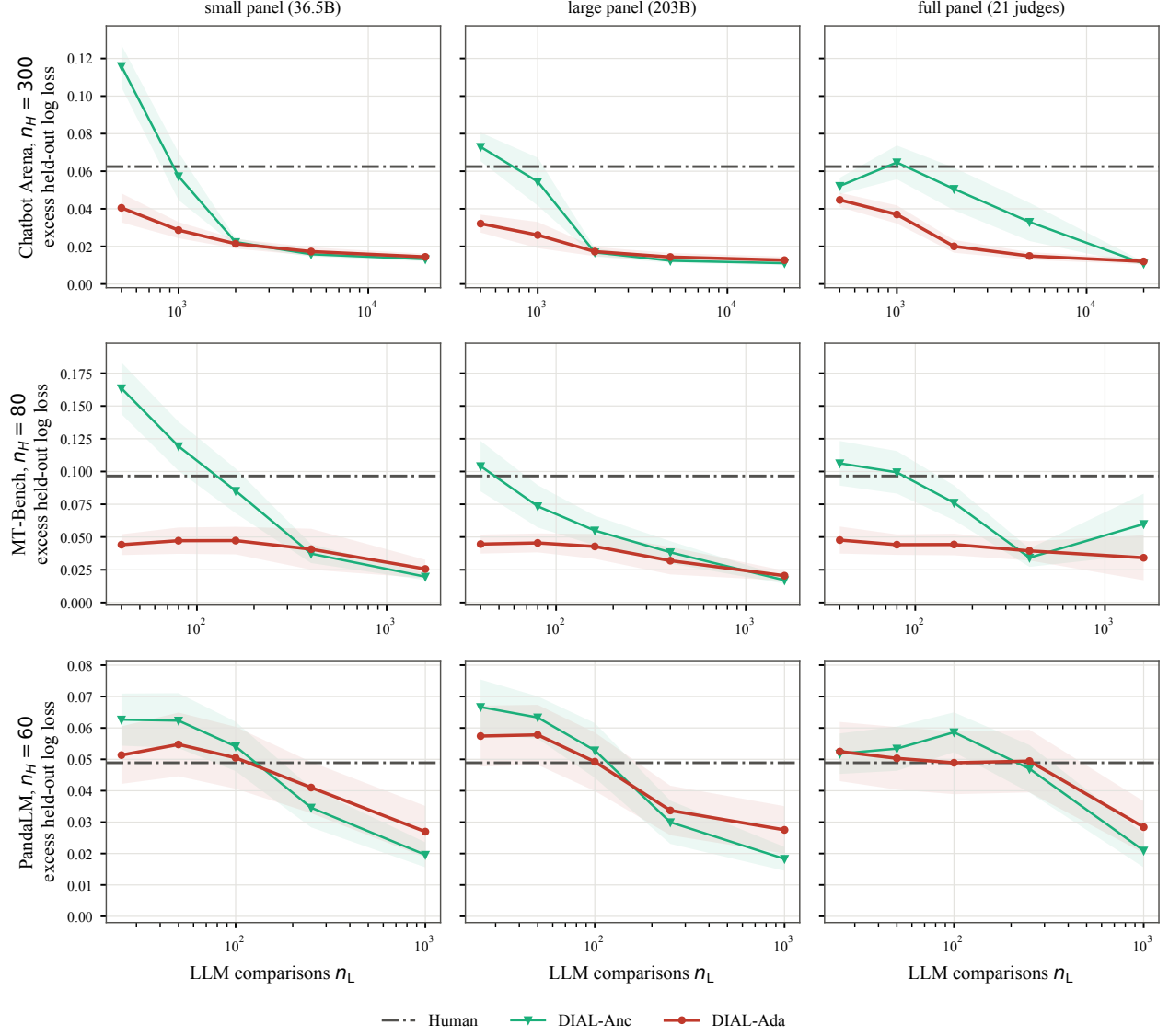


Figure G8: Complementary LLM-budget sweep for the small, large, and full judge panels of Figure G7. The human calibration budget is fixed at $n_H = 300, 80$, and 60 on Chatbot Arena, MT-Bench, and PandaLM, respectively, while balanced LLM training judgments are subsampled uniformly to n_L rows. Curves compare DIAL-Ada, at the GACV-selected weight, with DIAL-Anc and Human. All curves use up to the same 50 record splits, with ± 1.96 Monte Carlo standard errors; the y -axis is shared within each row.